

Systematic Review

CO₂ Enrichment in Protected Agriculture: A Systematic Review of Greenhouses, Controlled Environment Systems, and Vertical Farms—Part 2

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Abstract: CO₂ enrichment in protected agriculture has been extensively studied as a strategy to enhance crop productivity, resource use efficiency, and climate resilience. This systematic review examines the scientific literature on CO₂ enrichment in greenhouses, vertical farms, and controlled environment agriculture (CEA) systems, with a focus on its impact on crop physiology, photosynthesis, agricultural yield, modeling and simulation techniques, injection technologies, and sustainability challenges. A comprehensive bibliometric and systematic search was conducted in the Scopus database using key terms related to CO₂ enrichment and sustainable protected agriculture, following the PRISMA methodology. From an initial set of 212 documents, 171 were selected after removing duplicates, inaccessible articles, and studies not directly relevant to this context. The findings indicate that CO₂ enrichment can significantly improve photosynthetic efficiency, water use efficiency, and crop productivity, although its impact varies depending on species, environmental conditions, and application strategies. Computational models, such as CFD and machine learning, have optimized CO₂ distribution in controlled environments, contributing to more precise and resource-efficient agricultural practices. However, environmental and economic concerns, particularly energy consumption, carbon footprint, and the sustainability of CO₂ sources, remain critical challenges. To ensure the sustainable adoption of CO₂ enrichment, it is essential to integrate renewable energy sources, carbon capture and reuse technologies, and advanced CO₂ injection systems. This review provides a holistic assessment of current knowledge, identifying opportunities and barriers for the development of climate-smart protected agriculture systems that align with global sustainability goals and contribute to food security and environmental stewardship.

Keywords: carbon footprint climate-smart agriculture; microclimate regulation; photosynthesis optimization; sustainable agricultural

1. Introduction

Accelerated population growth and economic development, along with the intensification of urbanization, have significantly increased pressure on infrastructure and natural resources to meet the global demand for food, water, and energy [1]. It is estimated that by the year 2050, global energy consumption will increase by 80%, while food demand will grow by 60% and water consumption will rise by 55% [2]. In this context, agriculture accounts for more than 70% of global freshwater withdrawals, highlighting the need to improve water resource use efficiency [3].

Greenhouses and Plant Factories With Artificial Lighting (PFAL) emerge as a key alternative to address these challenges, as they can achieve yields equivalent to traditional agriculture while using only 10% of the water required in open-field cultivation [3]. However, this type of infrastructure is among the highest energy consumers in the agricultural sector. In a protected production system, the interactions between water, energy, and food resources, known as the Water–Energy–Food (WEF) Nexus, are particularly evident [4]. This underscores the need for strategies to optimize the use of these inputs. To enhance the sustainability of greenhouse and PFAL production, it is imperative to develop innovative solutions that reduce high energy demand without compromising yield [5]. These strategies must take into account not only water consumption efficiency but also the provision of optimal environmental conditions for crop growth and development [6].

Within the context of microclimate management in these production technologies, carbon dioxide (CO₂) enrichment has become a relevant strategy in protected agriculture due to its potential to enhance resource use efficiency and increase agricultural productivity [7]. The physiological response to CO₂ varies depending on the species and cultivation conditions, influencing the photosynthetic rate, water use efficiency (WUE), and biomass partitioning [8]. In horticultural crops, such as tomato (*Solanum lycopersicum*), pepper (*Cap-sicum annuum*), and basil (*Ocimum basilicum*), elevated CO₂ concentrations have increased biomass by up to 50%, promoting the development of both photosynthetic and reproductive organs [9]. In high-tech systems, such as Plant Factories With Artificial Lighting (PFAL), the combination of CO₂ enrichment with artificial lighting and hydroponic techniques has enabled an increase in production density without compromising product quality. However, its efficiency depends on the precise control of multiple environmental variables, including CO₂ concentration, light intensity and spectrum, temperature, humidity, and nutrient solution composition. These factors must be carefully managed to optimize photosynthesis, ensure proper plant development, and maximize resource use efficiency in controlled environments [10].

However, the implementation of CO₂ enrichment requires advanced strategies to optimize its efficiency and minimize losses due to ventilation. The use of simulation models and computational fluid dynamics (CFD) has been fundamental in assessing gas distribution within greenhouses and improving its application [11,12]. Injection technologies, such as automated dosing, industrial emissions capture, and the use of renewable sources, have emerged as alternatives to reduce costs and enhance the sustainability of the process [13]. However, economic viability remains a critical factor, as operational costs must be justified by improvements in both productivity and crop quality [14]. Additionally, the efficiency of CO₂ enrichment varies depending on the cultivated species, the phenological stage, and

interactions with other climatic factors, requiring a comprehensive approach for its optimal application in different agricultural systems [15].

On the other hand, systematic reviews play a fundamental role in identifying and synthesizing existing evidence on CO₂ enrichment in protected agriculture, enabling a rigorous assessment of its benefits and limitations [16]. This methodological approach enables the consolidation of information from previous studies using standardized criteria, facilitating the formulation of strategies based on scientific evidence [17]. Unlike narrative reviews, which provide a general overview without strict methodological criteria, systematic reviews follow a structured protocol to identify, select, analyze, and synthesize relevant information [18]. This will enable a quantitative and qualitative assessment of the impact of CO₂ on different crops and production systems. Ultimately, this type of review provides a comprehensive perspective on current research trends, helping to identify opportunities and outstanding technical challenges [19].

Based on the above, this review article is structured around key questions that guide the analysis of the existing literature on the impact of CO₂ enrichment in protected agricultural systems. Specifically, it aims to address the following questions. How does CO₂ enrichment influence crop physiology and photosynthesis? What is the effect of CO₂ enrichment on crop production and yield? What models and simulation techniques have been used to analyze CO₂ distribution in controlled environments and its impact on plant development? What are the most efficient technologies for CO₂ injection and enrichment in greenhouses? What role do Plant Factories with Artificial Lighting (PFAL) play in agricultural production with supplemental CO₂? What are the economic benefits and energy challenges associated with this strategy? What environmental and sustainability implications are related to CO₂ use in protected agriculture?

Although CO₂ enrichment has been extensively studied, the novelty of this review lies in its comprehensive and integrative approach, which combines experimental findings, modeling techniques, technological advancements, and economic–environmental assessments in a single framework. Unlike previous studies that focus primarily on physiological or agronomic responses, this review provides a multidimensional perspective, incorporating aspects like spatial CO₂ distribution modeling, advanced injection technologies, and the role of emerging controlled environment agriculture systems (e.g., PFALs). By bridging these different dimensions, this study identifies critical knowledge gaps and proposes a roadmap for future research and the practical implementation of CO₂ enrichment strategies in modern agricultural systems.

The objective of this systematic review is to provide a comprehensive and up-to-date analysis of the impact of CO₂ enrichment in protected agriculture. It aims to integrate findings from experimental and modeling studies, identifying trends, advantages, limitations, and challenges in its application. Additionally, economic, technological, and environmental aspects will be discussed to offer recommendations for the efficient implementation of CO₂ enrichment strategies in various agricultural production systems. Finally, future research directions will be proposed to contribute to the sustainable development of more efficient and resilient agricultural systems in the face of climate change.

2. Materials and Methods

The systematic review was conducted following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology, ensuring a transparent and reproducible process in the identification, selection, and analysis of the literature on CO₂ enrichment in protected agriculture [20,21]. To address the key questions formulated in this study, a systematic literature review was conducted following a structured methodology that ensured the comprehensiveness and reproducibility of the process (Figure 1). In

addition, a checklist following the updated PRISMA 2020 guidelines is included in the Supplementary Material.

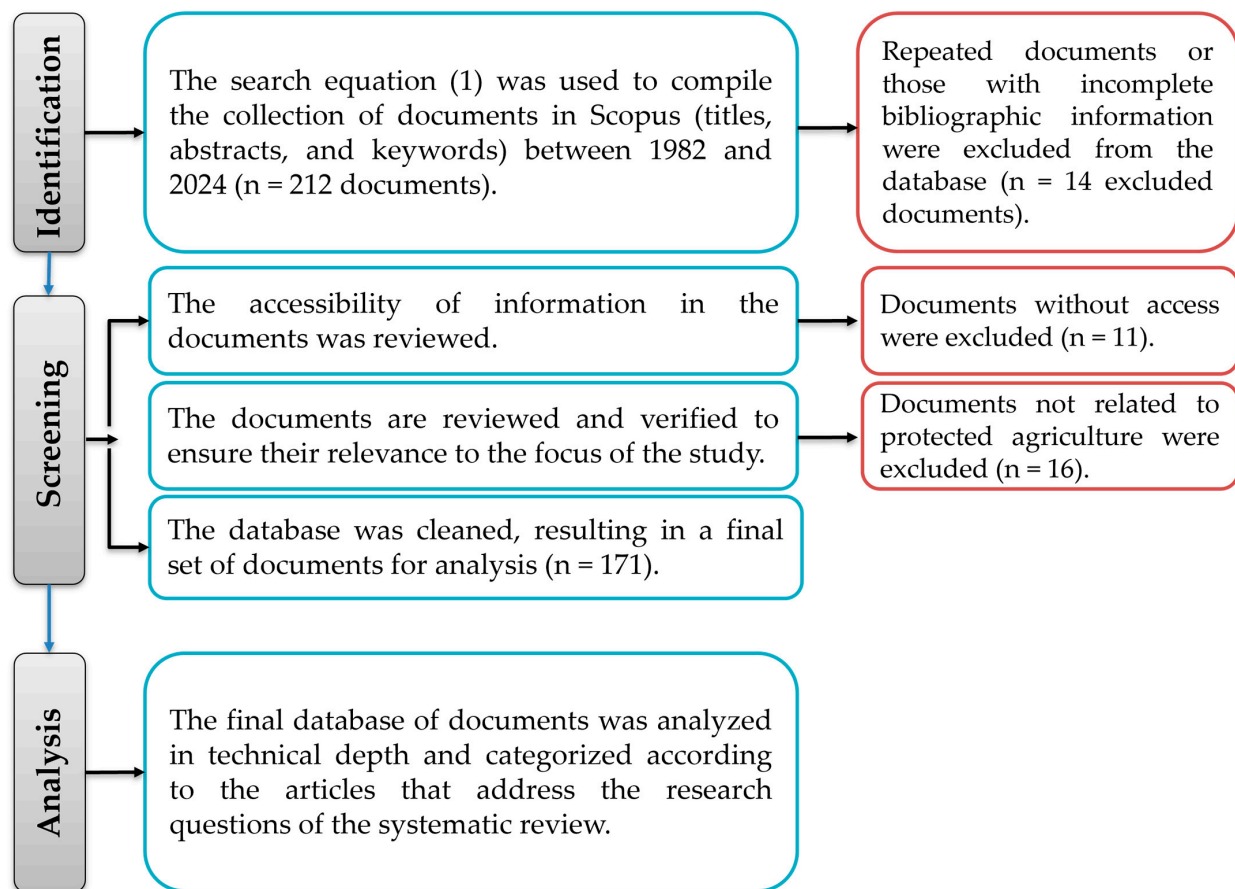


Figure 1. Methodological scheme of the systematic review.

2.1. Identification

A systematic search was conducted in Scopus using the following search equation:

(TITLE-ABS-KEY ((“CO₂ enrichment” OR “carbon dioxide enrichment” OR “CO₂ injection” OR “carbon dioxide injection”) AND (“greenhouse” OR “vertical farm” OR “plant factory” OR “controlled environment agriculture” OR “CEA” OR “protected agriculture”) AND (“plant growth” OR “crop yield” OR “photosynthesis” OR “agricultural production”)) AND NOT TITLE-ABS-KEY (“climate change” OR “wild ecosystems” OR “ocean” OR “forests”)).

This strategy enabled the collection of documents published between 1982 and 2024, resulting in a total of 212 documents.

2.2. Screening

The selection process was carried out in multiple stages following PRISMA guidelines. First, 14 duplicate documents or those with incomplete bibliographic information were removed. Then, 11 documents without full-text access were excluded. Subsequently, a thematic focus review was conducted by analyzing the abstracts and content of the articles to verify their relevance within the context of protected agriculture, leading to the exclusion of 16 documents that did not meet the selection criteria. After this refinement, the final database consisted of 171 documents for analysis.

2.3. Analysis

The selected articles were classified and analyzed based on the key questions established in the introduction. To achieve this, the following analytical categories were structured.

Impact of CO₂ Enrichment on Crop Physiology and Photosynthesis: Studies were analyzed to evaluate how CO₂ supplementation affects the photosynthetic rate, stomatal conductance, carbon assimilation, and water use efficiency in various species cultivated in greenhouses and plant factories.

Effect of CO₂ Enrichment on Crop Production and Yield: Data were collected on biomass increase, fruit production, and nutritional quality of horticultural, fruit, and ornamental crops under carbon enrichment.

Modeling and Simulation of CO₂ Distribution in Protected Environments: Studies employing computational fluid dynamics (CFD) models, physiological models, and machine learning techniques were reviewed to optimize CO₂ distribution and absorption in greenhouses and hydroponic systems.

CO₂ Injection and Enrichment Technologies in Greenhouses: Innovative methods for controlled CO₂ application were identified, including the use of exhaust gases, biogas, membrane systems, and liquefied CO₂.

Role of Plant Factories (PFAL) in CO₂-Supplemented Production: Research on the use of CO₂ in vertical farming was explored, assessing its impact on photosynthetic efficiency, product quality, and economic viability.

Economic Benefits and Energy Challenges of CO₂ Enrichment: Economic evaluations and life cycle analyses were reviewed to determine the profitability of CO₂ enrichment in different production contexts.

Environmental and Sustainability Implications: The effects of supplemental CO₂ on carbon footprint, energy, and water consumption and its potential to enhance the climate resilience of protected agricultural systems were analyzed.

Each analytical category was addressed by integrating the most relevant findings from the literature, highlighting knowledge gaps and opportunities for future research in the field. Based on the collected information, a qualitative synthesis of the results was conducted, identifying emerging trends and outstanding challenges in the implementation of CO₂ enrichment in protected agriculture. The integration of studies enabled the consolidation of conclusions regarding the efficiency of this strategy, its impact on agricultural production, and its long-term sustainability.

3. Results and Discussion

3.1. Impact of CO₂ Enrichment on Crop Physiology and Photosynthesis

CO₂ enrichment has significant effects on the physiological and biochemical processes of plants, including alterations in the photosynthetic rate, gas exchange, and nitrogen metabolism, which directly influence growth, productivity, and crop quality. The studies discussed in this section examine the impact of CO₂ on various horticultural and fruit crops, as well as on ornamental and forest species, highlighting the adaptive responses of plants under elevated CO₂ concentrations. Understanding these responses is essential for optimizing agricultural production in controlled environments, improving resource use efficiency, and enhancing crop quality.

Tomato (*Solanum lycopersicum* L.) is one of the most extensively studied greenhouse crops due to its economic importance and its response to carbon dioxide enrichment [22]. Photosynthesis, a key process in converting light energy into essential compounds for growth, is fundamental to agricultural productivity. Optimizing this process through the management of factors like CO₂ is crucial for maximizing yield and sustainability [23].

CO₂ concentration, along with other microclimatic conditions, significantly influences the growth and yield of tomatoes. This analysis compiles recent studies on the effects of CO₂ on the photosynthesis and physiology of this crop.

The adequate supply of CO₂, within a range of 600 to 1000 $\mu\text{mol mol}^{-1}$, has been shown to enhance photosynthetic indices at all developmental stages of this crop, increasing chlorophyll and carotenoid content. These increases are closely related to improved photosynthetic efficiency, which significantly contributes to biomass accumulation and the optimal development of photosynthetically active organs, directly impacting fruit yield and quality. In an experiment conducted with the cultivar “Xinghai 12”, the net photosynthetic rate increased significantly, reaching its peak at a CO₂ concentration of 1000 $\mu\text{mol mol}^{-1}$. Additionally, a decrease in stomatal conductance was observed, leading to improved water use efficiency [24]. Another study demonstrated that continuous CO₂ enrichment increased both the fresh and dry weight of the plants, enhancing overall crop yield [25].

An additional study analyzed photosynthetic activity, chlorophyll α (Chl- α) fluorescence, and carbohydrate content in tomato plants grown in greenhouses under CO₂ enrichment and supplemental lighting. During winter, the results showed a decrease in photosynthetic activity due to a limited photosynthetic photon flux (PPF), although no evidence of photoinhibition was observed. In spring, an increase in PPF was reported, leading to sucrose and starch accumulation due to CO₂ enrichment. However, there was also a downregulation of photosynthesis caused by sink limitation. The ratio of variable fluorescence to maximum fluorescence $F(v)/F(m)$ decreased, indicating reversible and adaptive inhibition in the leaves, particularly in March. These findings highlight how tomato plants adjust their photosynthetic physiology in response to seasonal variations in PPF and CO₂ levels [26].

The effect of CO₂ concentration also depends on its interaction with other microclimatic factors, such as light and temperature. For example, under low irradiance conditions in winter, CO₂ can significantly increase the net photosynthetic rate, compensating for the limited availability of light. Additionally, tomato plants adjust their light compensation point in response to seasonal changes, allowing for photosynthetic optimization even under variable climatic conditions. The interaction between CO₂ and temperature also influences key processes, such as Rubisco activity, which operates efficiently at optimal temperatures, maximizing carbon fixation and, consequently, biomass accumulation. Studies conducted under winter conditions, when irradiance is low, showed that CO₂ enrichment up to 1000 $\mu\text{mol mol}^{-1}$ increased net photosynthesis by 51% and water use efficiency by 60%, effectively offsetting light limitations [27]. Additionally, during the transition from winter to spring, tomato leaves adjusted their light compensation point, enhancing photosynthesis and transpiration due to increased CO₂ levels and greater irradiance [28].

The combination of supplemental lighting and CO₂ enrichment has shown synergistic effects on biomass accumulation and yield. In autumn-to-spring production cycles, the use of supplemental lighting at 200 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ and CO₂ enrichment at 800 $\mu\text{mol mol}^{-1}$ resulted in a 36.6% increase in fruit weight and a 60.8% increase in yield per plant [29]. Similarly, the use of diffuse covers combined with CO₂ concentrations of 800 $\mu\text{mol mol}^{-1}$ increased annual tomato yields by 30%, demonstrating the importance of optimizing the canopy microclimate to maximize productivity [30].

CO₂ enrichment plays a crucial role under abiotic stress conditions, such as salinity and reduced irrigation. The application of elevated CO₂ levels at 700–800 $\mu\text{mol mol}^{-1}$ mitigated the negative effects of salinity, improving photosynthesis and biomass accumulation, particularly when combined with appropriate nitrate-to-ammonium ratios [31]. Under reduced irrigation conditions, the combination of photosynthetic bacteria and elevated CO₂ enhanced nutrient absorption efficiency, improving both yield and fruit quality [32]. On the

other hand, in semi-closed greenhouses, where light transmission is often reduced, CO₂ enrichment can compensate for this limitation. For example, a 10% decrease in light intensity can be offset by increasing the CO₂ concentration by approximately 100 $\mu\text{mol mol}^{-1}$ and even overcompensated by up to 40% when maintaining CO₂ levels at 1000 $\mu\text{mol mol}^{-1}$ [27].

Studies conducted during the summer demonstrated that elevated CO₂ levels up to 1000 $\mu\text{mol mol}^{-1}$ combined with cooling using a reflective NIR film proved to be an effective method for maintaining tomato productivity and quality, thereby reducing heat stress. In a similar study, tomato plants exposed to elevated CO₂ (800 $\mu\text{mol mol}^{-1}$) and thermal stress at 42 °C showed significant improvements in photosynthesis and post-stress recovery compared to plants grown under normal CO₂ conditions. Under heat stress, CO₂ enrichment increased the net photosynthetic rate (P_n), maximum carboxylation capacity (V_{max}), and RuBP regeneration (J_{max}), while also reducing photoinhibition and damage to photosystems. Additionally, improvements in redox homeostasis were observed, evidenced by higher ASA:DHA ratios (ascorbate (ASA) and dehydroascorbate (DHA)), higher GSH:GSSG ratios (reduced glutathione (GSH) and oxidized glutathione (GSSG)), and increased NADP⁺ concentrations during recovery. These findings highlight the potential of elevated CO₂ to protect and restore the photosynthetic functionality of tomatoes under extreme heat stress conditions [33].

From a physiological perspective, it is important to highlight that carbohydrate accumulation in source leaves, induced by elevated light and CO₂ levels, can limit long-term photosynthetic efficiency by inhibiting the regeneration of electron acceptors in the Calvin cycle, leading to metabolic saturation. This effect can be mitigated through appropriate management of light intensity and CO₂ supply, as well as practices that promote the redistribution of accumulated carbohydrates to storage organs, ensuring an efficient metabolic balance. In experiments with exogenous glucose feeding, a decrease in the maximum photosynthetic rate and foliar pigment content was observed, underscoring the importance of properly managing light and CO₂ conditions to prevent counterproductive effects [34].

Regarding the genetic analysis of tomato plants, key genes, such as *Soly720*, have been identified, whose overexpression significantly enhanced chlorophyll content, photosynthetic characteristics, and key enzymatic activities under high CO₂ concentration conditions [35]. Additionally, strategies like the optimal control of CO₂ concentration levels have proven to be crucial not only for maximizing yield but also for optimizing the net photosynthesis of the crop. Mathematical models used to estimate net photosynthesis revealed that continuously adjusting CO₂ concentration maximizes photosynthetic efficiency, particularly when using pure CO₂ or exhaust gases with heat storage, which also increases financial margins [15].

In other studies conducted in Japan and Thailand, CO₂ enrichment at levels of 600–800 $\mu\text{mol mol}^{-1}$ resulted in a 19.32% increase in biomass, a photosynthetic rate of 34.4 $\mu\text{mol m}^{-2} \text{s}^{-1}$, and total soluble solids in the fruit reaching 4.9 °Brix compared to plants grown under ambient CO₂ conditions [36]. These results demonstrate that CO₂ enrichment enhances not only photosynthesis but also the nutritional quality of the fruit, making it an essential practice for greenhouse cultivation during winter.

In the context of sweet pepper (*Capsicum annuum* L.) cultivation, CO₂ enrichment in protected environments has shown significant improvements in plant physiology, maximizing yield and resource use efficiency, particularly water. However, these benefits are modulated by factors like photosynthetic acclimation, irrigation conditions, and changes in nutrient composition, requiring precise management to optimize outcomes [37]. CO₂ enrichment in Mediterranean greenhouses has been shown to increase the net photosynthetic rate, particularly during the early growth stages of the crop [38].

Studies have revealed that CO₂ concentrations between 600 and 1000 µmol mol⁻¹ significantly increased fruit fresh mass and water use efficiency, particularly under limited irrigation conditions. This effect is attributed to greater efficiency of the photosynthetic apparatus in carbon capture under high CO₂ concentrations, enhancing assimilation rates and biomass accumulation. However, the relationship between CO₂ and yield was found to be nonlinear, reaching an optimal point at 600 µmol mol⁻¹ under adequate irrigation conditions [39]. On the other hand, the curing and acclimatization of grafted pepper transplants are essential for their optimal development, directly influencing their physiology and photosynthetic capacity. A study demonstrated that CO₂ enrichment at 1000 µmol mol⁻¹ combined with higher photosynthetic photon flux (PPF) levels of 150 µmol m⁻²s⁻¹ significantly increased CO₂ exchange rates, particularly under low light intensity. This enhancement in photosynthesis led to improved plant growth, reflected in greater dry weight, larger leaf area, and higher SPAD values. Additionally, higher PPF levels strengthened leaf anatomical structures, resulting in denser tissues and optimized chloroplasts [40].

Prolonged exposure to high CO₂ concentrations can induce photosynthetic acclimation, a process characterized by a reduction in photosynthetic capacity due to negative feedback mechanisms, such as the accumulation of non-structural carbohydrates. This phenomenon became evident after 135 days of enrichment in a study conducted in Mediterranean greenhouses, where sweet pepper plants showed a progressive decline in net photosynthetic rate. The photosynthetic response was more closely related to foliar nitrogen concentration than accumulated carbohydrates, suggesting that nutritional management, specifically the use of NH₄⁺ instead of NO₃⁻, may mitigate acclimation and extend the benefits of CO₂ enrichment [37].

Similarly, another study demonstrated that increasing the relative humidity to 60% in combination with CO₂ enrichment was effective in improving the photosynthetic rate during morning hours. This effect is attributed to the fact that 60% relative humidity increases stomatal conductance, facilitating CO₂ uptake by the leaves and maximizing photosynthetic efficiency. However, in the afternoon, when stomatal conductance naturally decreases, the effect of humidification was marginal, even under CO₂ enrichment conditions. This finding highlights the importance of scheduling CO₂ management and irrigation strategies according to the circadian rhythms of the plants [41]. In warm climate regions, such as the Mediterranean, CO₂ enrichment also benefits crop production by increasing the optimal growth temperature for C3 plants like sweet pepper. This adjustment allows for prolonged CO₂ application periods by reducing the need for greenhouse ventilation. Studies have reported a 27% increase in plant yield under CO₂ enrichment in high-temperature conditions, highlighting the crop's physiological adaptability to environmental management [42].

Finally, the efficient management of CO₂ enrichment in sweet pepper cultivation depends not only on climatic conditions and ventilation levels but also nutrient management and salinity control. For example, the combination of NH₄⁺ in nutrient solutions under low salinity conditions delayed photosynthetic acclimation and extended the benefits of CO₂ enrichment. This integrated approach, which combines nutritional management with CO₂ supply strategies, is crucial for maximizing both the economic and biological efficiency of the crop [37].

In cucumber (*Cucumis sativus*) cultivation, CO₂ enrichment combined with supplemental lighting (SL + CO₂) significantly enhanced crop yield. A study conducted in a plastic greenhouse demonstrated that this treatment increased the yield by 35% compared to a control without supplementation. Additionally, an increase in the number of female flowers and dry matter content in the fruits was observed. This suggests that elevated CO₂ stimulates photosynthetic activity, leading to greater assimilated production and distribution of reproductive organs, ultimately increasing the final yield [43].

In another study, increasing the CO₂ concentration to 1200 µmol mol⁻¹ significantly enhanced the photosynthetic rate of cucumber, with a 58% increase during the seedling stage and a 74% increase during the early fruiting stage, compared to plants grown under an ambient CO₂ concentration of 380 µmol mol⁻¹. This increase in photosynthesis resulted in a 130% rise in sugars released in root exudates during the seedling stage and a 102% increase during the fruiting stage. The total sugar content correlated with increased root mass, suggesting that CO₂ enrichment not only improves photosynthetic activity but also alters the composition and release of photosynthates toward the roots [44].

Under summer conditions, CO₂ enrichment at levels of 800–1000 µmol mol⁻¹, combined with day–night temperature fluctuations (DNFs), significantly improved the distribution of dry matter to cucumber fruits, with a 15% increase compared to ambient CO₂ conditions. Low night temperatures (LTs) inhibited respiration, promoting a 30% increase in dry matter accumulation under elevated CO₂ conditions. Additionally, fruit fresh weight increased by 8–12% under CO₂ enrichment in the DNF treatments. This increase in the accumulation and distribution of photosynthates highlights the ability of elevated CO₂ to enhance fruit quality and photosynthetic efficiency by adjusting metabolic rates and reducing respiratory losses [45].

In solar greenhouses in northern China, CO₂ enrichment applied in the morning and afternoon significantly increased the photosynthetic rate and dry matter production in cucumber. During the fruiting stage, photosynthetic rates were 110 to 175% higher in enriched plants. Plants receiving CO₂ during both periods produced 68% more dry matter than those enriched only in the morning and 113% more than non-enriched plants. These results highlight the importance of applying CO₂ at specific times to optimize photosynthetic efficiency and growth in protected systems [46].

In lettuce (*Lactuca sativa* L.) cultivation, the use of overnight supplemental lighting (OSL) with red light-emitting diodes (LEDs) has been shown to increase dry matter accumulation by extending the duration of active photosynthesis. However, this strategy can induce photosynthetic stress due to excessive photosynthetic activity, as evidenced by a 10% reduction in the maximum photosynthetic rate and the quantum yield of photosystem II (Fv/Fm). Additionally, CO₂ enrichment at a concentration of 1000 ppm significantly reduced this stress, restoring optimal levels of photosynthesis and Fv/Fm. This highlights the ability of CO₂ to mitigate the negative impact of prolonged lighting, providing an effective tool for managing plant physiology under high light demand conditions [47].

Another study developed a model integrating the effects of temperature, photosynthetic photon flux density (PPFD), and CO₂ concentration on lettuce photosynthesis. The results indicated that CO₂ promotes photosynthesis within an optimal range of conditions, but, beyond this point, its beneficial effects diminish. The optimal combination identified included a PPFD of 897.3 µmol m⁻² s⁻¹, a temperature of 28.9 °C, and a CO₂ concentration of 2160 µmol mol⁻¹, achieving a maximum net photosynthetic rate of 36.0 µmol m⁻² s⁻¹. This quantitative approach allows for the optimization of a protected environment design, maximizing photosynthetic efficiency while minimizing resource waste [48].

In other crops, increasing the CO₂ concentration to levels between 700 and 1200 µmol mol⁻¹ has been shown to enhance the net photosynthetic rate in fruit crops, such as strawberry (*Fragaria ananassa* Duch.), oriental melon (*Cucumis melo* L.), and grapevine (*Vitis vinifera* L.). In strawberries, for instance, CO₂ enrichment increased the photosynthetic rate by 129.7% and the intercellular CO₂ by 43.7%, reflecting an improvement in carbon assimilation capacity under controlled conditions [49]. Similarly, in oriental melon, a significant increase was observed in the activity of key photosynthetic enzymes, such as Rubisco and FBPase, which supported enhanced carbon assimilation and yield [50].

Exposure to elevated CO₂ concentrations induces physiological changes that enhance photosynthetic efficiency, such as an increased light saturation point and a reduced light compensation point. In strawberries, these effects were accompanied by the upregulation of genes involved in photosynthesis, enabling greater transport of photosynthetic products and accelerated leaf growth [51]. In grapevine, elevated CO₂ enhanced the electron transport rate and carboxylation efficiency, reaching peak values in the net photosynthetic rate and apparent quantum efficiency at concentrations of 700 ppm. However, at levels above 1000 ppm, a progressive inhibition of photosynthesis was observed due to negative feedback mechanisms [52].

In mango (*Mangifera indica* L.), the influence of the vertical position of CO₂ enrichment on photosynthesis in greenhouse-grown plants was evaluated using a three-dimensional plant model and ray-tracing simulations. Sensors were installed to measure CO₂ concentration distributions under four treatments: no enrichment (control) and enrichment at 0.5 m (E0.5), 1.0 m (E1.0), and 1.5 m (E1.5). Leaf photosynthetic rates were estimated using the Farquhar, von Caemmerer, and Berry model. The results showed that CO₂ concentrations were higher in the enrichment treatments compared to the control, with increases of 45.6% (E0.5), 48.4% (E1.0), and 153.6% (E1.5). CO₂ consumption per plant also increased, being 50.0%, 63.6%, and 49.8% higher, respectively, compared to the control (1.76 g CO₂·h^{−1}) [53]. These increases were associated with improved CO₂ distribution within the tree canopy and greater efficiency in light interception, both of which are key factors for maximizing the photosynthetic rate.

In fruit crops, the photosynthetic response to elevated CO₂ is also highly influenced by factors like light and temperature. In strawberries, the effect of CO₂ was more pronounced under high light intensity and temperatures close to 30 °C, indicating that enrichment is most effective when controlled ventilation is maintained to reduce CO₂ and heat loss [54]. Another study evaluated an automatic CO₂ concentration control method in greenhouses to optimize strawberry productivity in northern Kyushu, Japan. It was reported that maintaining CO₂ levels above 800 µmol mol^{−1} without ventilation and 400 µmol mol^{−1} with ventilation (Figure 2) increased yields by 25% and improved the Brix degree of the fruits by 0.2 to 1.2%. Additionally, the crop's photosynthetic rate increased linearly with CO₂ concentrations up to 800 µmol mol^{−1}, promoting the translocation of photoassimilates to the fruits, the primary sink, resulting in more efficient resource distribution and higher yields [55].

Similarly, in melon, CO₂ concentrations between 800 and 1200 µmol mol^{−1} improved photosynthetic enzyme activity and dry matter accumulation. However, salinity significantly reduced these benefits by negatively affecting stomatal conductance and leaf surface area [56]. Water and nutrient management also interact with elevated CO₂ to regulate the physiology of fruit crops. In watermelon, CO₂ enrichment mitigated the negative effects of low nitrogen supply by increasing the photosynthetic rate and dry matter accumulation. Additionally, optimal irrigation at 120% of potential evaporation (Ep) was found to maximize the physiological benefits of elevated CO₂, highlighting the importance of precise resource management to optimize photosynthesis [57].

For common bean (*Phaseolus vulgaris*), an increase in CO₂ concentration to 700 µL l^{−1} significantly enhanced photosynthesis (+73% at 40 days after germination) and improved key physiological traits, such as leaf area (+36%) and leaf mass (+57%). Additionally, soluble (+49%) and insoluble (+64%) sugar contents in the leaves increased, along with a higher C/N ratio (+19%), although a reduction in leaf nitrogen content (−15%) was observed. At the production level, plants exposed to elevated CO₂ showed higher fresh (+44%) and dry (+110%) pod weights, as well as an increase in individual dry pod weight (+100%), although with a slight reduction in the number of seeds per pod (−9%) [58].

Photosynthesis is a fundamental process in cut flowers and ornamental plants, as it directly influences biomass accumulation, floral development, and the efficient use of resources, such as water and light [59,60]. In high-value ornamental crops, such as *Phalaenopsis Queen Beer* (*Phalaenopsis*), *Gerbera* (*Gerbera jamesonii*), and *Rose* (*Rosa hybrida*), CO₂ enrichment has proven to be an effective strategy for optimizing productivity and quality, especially when combined with precise light and temperature management.

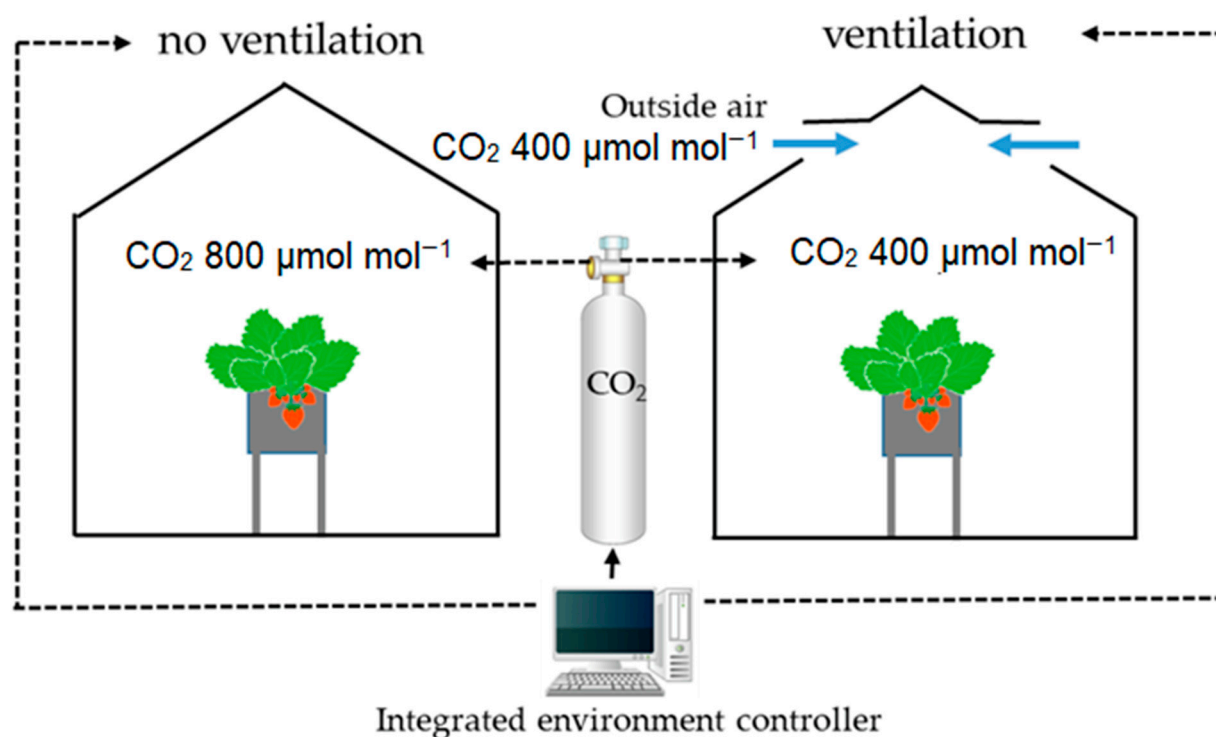


Figure 2. Schematic illustration of the treatments in strawberry. Figure taken from Tagawa et al. [55].

On the other hand, CO₂ enrichment combined with high light intensity significantly improves the photosynthetic rate and flower quality in species like *Phalaenopsis Queen Beer*. Plants grown under 1200 µmol mol⁻¹ of CO₂ and a high light intensity of $260 \pm 40 \mu\text{mol m}^{-2} \text{s}^{-1}$ exhibited a greater net CO₂ assimilation rate and accelerated floral bud development compared to plants grown under low light conditions and 400 µmol mol⁻¹ of CO₂. Additionally, improvements were observed in spike length and total flower count, highlighting the importance of integrating CO₂ management with appropriate lighting conditions to maximize productivity [61].

In *Gerbera*, CO₂ enrichment increased the net photosynthetic rate by 52–66%, promoting the accumulation of photoassimilates, such as soluble sugars and starch. These physiological changes contributed to a 32–40% increase in plant dry weight. Additionally, flower quality improved, with an increase in flower number, size, anthocyanin concentration, and vase life. Although no significant differences were observed between daytime and morning CO₂ enrichment, the morning treatment proved to be more cost-effective, highlighting its potential for commercial production [62].

The photosynthetic responses of ornamental plants, such as *Calibrachoa*, *Petunia*, *Verbena*, *Sunflower*, *Pepper*, and *Geranium*, vary depending on the combination of irradiance, CO₂ concentration, and temperature. For example, in *Geranium* and *Sunflower*, it was shown that the net photosynthetic rate can be modeled as a function of these factors, allowing for the identification of optimal conditions to maximize photosynthesis. At elevated CO₂ concentrations, photosynthetic rates equivalent to different combinations of irradiance and

temperature were achieved, providing growers with options to reduce operational costs without compromising productivity [63].

In *Rosa hybrida*, CO₂ enrichment improved carbon fixation and its export to developing organs. During floral bud development, a significant increase was observed in the proportion of carbon exported from the leaves to the flowers, especially under saturating irradiance and CO₂ concentrations of 90 Pa. This mechanism ensures greater biomass accumulation in reproductive organs, enhancing both the size and quality of the flowers. Additionally, CO₂ enrichment increased the efficiency of carbon translocation, optimizing the use of photosynthetic resources [64].

In roses grown under high temperatures and photosynthetic photon flux density (PPFD), CO₂ enrichment increased the photosynthetic rate by more than 100% without compromising quantum efficiency or increasing the risk of photodamage. This effect was also reflected in a higher concentration of sugars in the leaves, indicating efficient use of fixed carbon for growth and floral development. Additionally, water use efficiency increased by 50%, highlighting the benefits of CO₂ enrichment for optimizing production under high light and thermal demand conditions [65,66].

3.2. Direct Effect of CO₂ Enrichment on Crop Production and Yield

The increase in atmospheric CO₂ in greenhouses and controlled production systems is directly related to the increase in yield and biomass of various crops [67,68]. The articles in this section explore how CO₂ enrichment enhances growth rate and fruit production, factors that are crucial for meeting food demand in the context of climate change [69]. The studies provide empirical data on the response of crops, such as tomato, cucumber, strawberry, and pepper, among others, offering scientific foundations for implementing sustainable agricultural practices that maximize productivity in greenhouses and protected cultivation systems.

CO₂ enrichment in greenhouses is a widely used strategy to increase agricultural productivity, particularly in C3 plants. Studies conducted at Aristotle University of Thessaloniki revealed that by elevating CO₂ levels to three times the environmental average of 410 ppm, an increase in the optimal growth temperature of tomatoes, cucumbers, and peppers was observed, ranging from 5 to 10 °C and reaching 30–32 °C. This adjustment allows for a delay in the onset of ventilation, thereby increasing the hours during the day when CO₂ enrichment can be applied, which also improved crop yield. In tomatoes, the average plant height increased by 8.64% and the central stem circumference by 7.76%. Cucumbers experienced rapid growth, with a 16.54% increase in fruit production, while peppers showed productivity increases of up to 17.5%. Additionally, as previously mentioned, energy savings were identified due to reduced ventilation, highlighting the potential of this strategy to optimize resources and increase energy efficiency [70].

In the case of tomato cultivation, agricultural yield depends on processes like the photosynthetic rate, light interception, and dry matter distribution. Research conducted in the Netherlands over the past 50 years has highlighted that increased light use efficiency (LUE) has been key to the overall increase in dry matter production. This advancement is attributed to the implementation of technologies like CO₂ enrichment, computerized environmental control, and soilless cultivation. Modern cultivars have shown better photosynthetic rates per plant and reduced light loss in the canopy, consistently increasing agricultural productivity [71].

In indoor cultivation systems, light-emitting diodes (LEDs) combined with CO₂ enrichment have proven highly effective for optimizing tomato seedling production. In a study analyzing four tomato cultivars—“Florida-47 R”, “Rebelski”, “Maxifort”, and “Shin Cheong Gang”—exposed to different CO₂ concentrations (448 ± 32, 1010 ± 48, and

$1568 \pm 129 \mu\text{mol mol}^{-1}$) and three levels of daily light integral (DLI) (6.5, 9.7, and $13 \text{ mol/m}^2/\text{day}$), it was reported that CO_2 enrichment between 1000 and $1600 \mu\text{mol mol}^{-1}$ increased light use efficiency by 38–44%, achieving significant increases in the net photosynthetic rate by up to 68% compared to the non-enriched CO_2 environment. Although CO_2 concentrations of $1600 \mu\text{mol mol}^{-1}$ did not increase fresh mass or stem elongation, optimal combinations, such as 13.0 DLI – 1000 CO_2 and 13.0 DLI – 1600 CO_2 , improved plant growth by 24–33% while maintaining stable energy consumption. This approach highlights the importance of adjusting both light conditions and CO_2 concentrations to maximize efficiency and sustainability in protected cultivation [72].

Strawberry has also shown positive results under CO_2 enrichment. In experiments conducted in Saudi Arabia with CO_2 levels of 600 ppm, the following increases were reported: photosynthetic rate by 129.7%, number of fruits per plant by 27.5%, and total yield by 42.2%. However, a reduction in the accumulation of nitrogen, phosphorus, and potassium in the leaves was observed, highlighting the need to adjust fertilization strategies in combination with CO_2 enrichment [49]. Similarly, in this crop, the implementation of the crop-localized CO_2 enrichment (CLC) system has allowed for an increase in CO_2 concentration in the crop canopy by 100 – $200 \mu\text{mol mol}^{-1}$.

This resulted in significant increases ranging from 10% to 26% in average fruit weight, 13% in the total number of fruits, and 22% in the total marketable yield compared to the conventional full-enrichment method. Furthermore, the use of the localized enrichment system stabilized CO_2 concentrations at 800 ppm during clear days, promoting photosynthesis and the transport of photosynthates. These results emphasize not only the need to optimize the duration, location, and intensity of enrichment but also to ensure reliable and, above all, sustainable outcomes (Figure 3) [73].

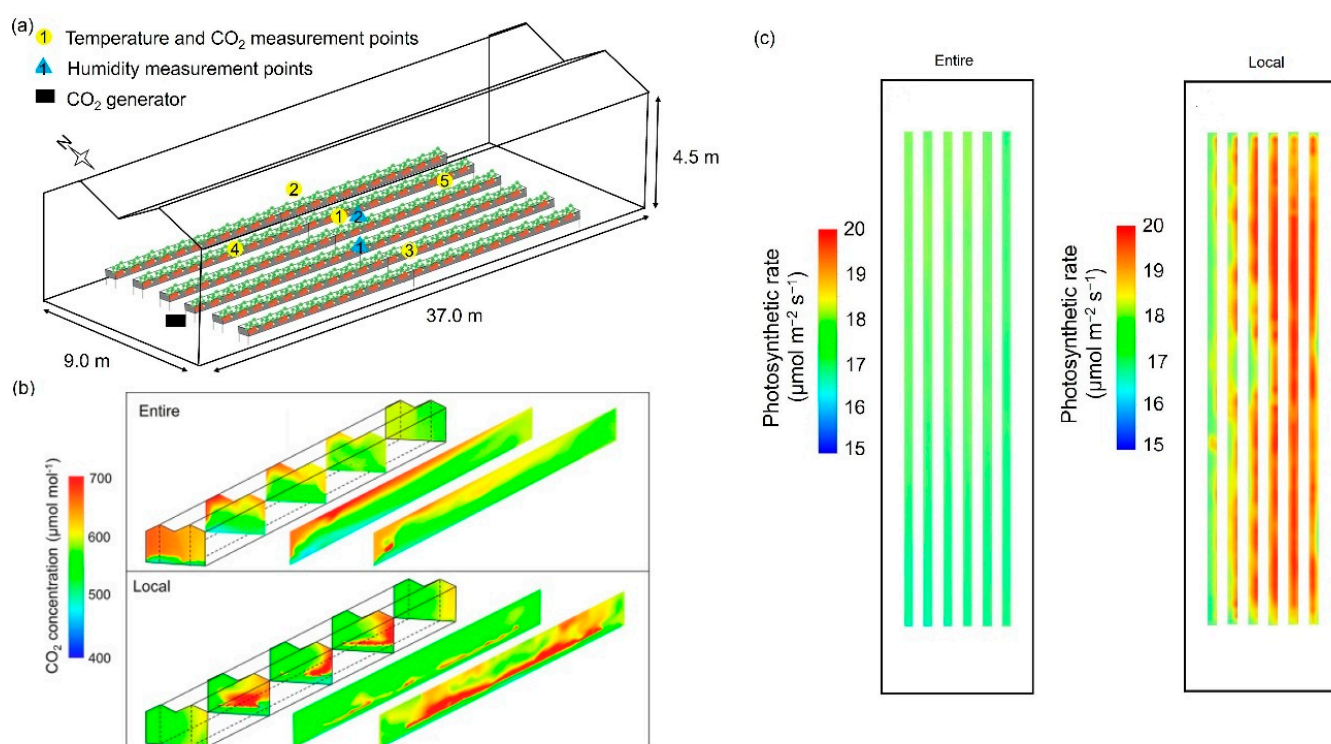


Figure 3. Comparison of a full CO_2 enrichment system with a localized one. (a) Analyzed greenhouse. (b) Photosynthetic rate inside of the greenhouse. (c) Photosynthetic rate in the crop canopy, calculated using CFD modeling. Figure adapted from Hidaka et al. [73].

In a study, changes in CO_2 concentrations and their impact on dry matter production in cucumber were investigated. The results showed that the maximum, minimum, and

average CO₂ concentrations varied throughout the seasons, with more severe depletion occurring during the hours before and after ventilation. This depletion was particularly notable in December, lasting for 4 h, while in March it extended to 8 h. CO₂ enrichment had a significant positive effect on photosynthesis and crop development. During the seedling stage, combined enrichment in the morning and afternoon at levels of $1100 \pm 100 \mu\text{L L}^{-1}$ increased dry matter production by 259% compared to plants without enrichment and by 68.08% compared to enrichment applied only in the morning [46].

CO₂ enrichment in a range of 600–1200 $\mu\text{mol mol}^{-1}$ significantly improved the growth and yield of bottle gourd cultivated in plastic houses. Increases were observed in plant height (3.90–19.48%), stem diameter (11.58–27.37%), leaf thickness (38.46–69.23%), and leaf area (26.09–49.38%). Additionally, the average fruit weight and total yield increased by 4.05–19.62% and 8.65–19.47%, respectively. The net photosynthetic rate and RuBP carboxylase activity reached their peak performance at a CO₂ concentration of 1000 $\mu\text{mol mol}^{-1}$, which was identified as optimal for the production of this crop in the spring [74].

In the case of peppers, the Italian cultivars “Palermo”, “Estar”, and “Charly”, grown in rock wool, were observed to have significantly increased plant height, stem diameter, leaf number, and leaf area under elevated CO₂ concentrations of 500 and 800 $\mu\text{L L}^{-1}$. Yield also increased notably, especially at 800 $\mu\text{L L}^{-1}$, with a higher average fruit weight. However, the incidence of blossom-end rot was higher in CO₂-enriched plants, being more pronounced at 500 $\mu\text{L L}^{-1}$. Additionally, enrichment improved water use efficiency for both marketable and total production, highlighting its potential to optimize resources in controlled systems [75].

In another study with the *Meite* cultivar, conducted under off-season conditions in phytotrons in the arid region of northwest China, combinations of supplemental light intensity (SLI) and CO₂ concentrations (400, 550, 700, and 900 $\mu\text{mol mol}^{-1}$) were evaluated. The results indicated that CO₂ enrichment significantly improved the net photosynthetic rate (Pn), water use efficiency (WUE), yield per plant (FYPP), and fruit storage quality. However, at higher CO₂ concentrations, a decrease in some nutritional attributes of the fruit was observed. Supplemental light intensity increased both the yield and fruit quality, with a 157% increase in FYPP. The optimal combination for maximizing yield and fruit quality was SLI combined with a moderate CO₂ concentration of 550 $\mu\text{mol mol}^{-1}$, resulting in high-quality fruits with excellent storage potential [76].

CO₂ enrichment has proven to be an effective strategy for improving the growth and development of lettuce. In an experiment conducted in solar greenhouses, three varieties of lettuce exposed to elevated CO₂ concentrations for 30 days showed a significant increase in chlorophyll content and vitamin C, while nitrate nitrogen levels decreased. Additionally, transcriptomic analysis revealed the positive regulation of genes related to photosynthesis, hormonal signaling, and carbohydrate metabolism, which enhanced the growth potential and nutritional quality of the crop [77]. Additionally, the use of CO₂ in drip irrigation proved to be a viable practice. In a study conducted with *Vera* lettuce, the application of CO₂ at rates of 1 and 3 L min^{-1} increased production by 26.5% and 14.5%, respectively, compared to the control group. This approach is not only efficient in terms of production but also economically viable, highlighting its potential for widespread implementation [78].

Exposure to different CO₂ concentrations (400, 800, 1200, and 1600 ppm) in controlled environment chambers revealed that levels between 400 and 800 ppm optimize the growth of *Rex* and *Rouxai* lettuces, improving both fresh and dry weight. In contrast, higher concentrations decreased photosynthetic efficiency and some nutritional components, such as violaxanthin, without significantly affecting key nutrients like lutein and anthocyanins. These results highlight the importance of maintaining optimal CO₂ levels to maximize the benefits of enrichment in controlled cultivation systems [79].

Finally, the cultivation of *Auvarona* lettuce under a hydroponic system using the nutrient film technique (NFT) showed a significant increase in fresh weight (24.7%) and dry weight (21.4%) with CO₂ supplementation at 800 ppm. A slight decrease in chlorophyll and nitrogen concentrations in the leaves was observed, but the overall yield improved, highlighting the viability of this practice in commercial hydroponic systems. These results reinforce the potential of CO₂ enrichment to boost productivity, although attention should be paid to possible variations in nutritional quality depending on the cultivated species [9].

For leafy crops like Bok choy (*Brassica chinensis*), elevated CO₂ levels significantly increased the biomass, especially under concentrations double that of the ambient environment (840 ppm). Additionally, this CO₂-enriched environment promoted nitrogen and potassium absorption while improving phosphorus use efficiency. However, a decrease in available soil nutrients was reported, emphasizing the importance of nutritional management strategies to maintain long-term productivity in these systems [80].

The use of CO₂ enrichment in medicinal plants, such as *Taxus baccata*, *Hypericum perforatum*, and *Echinacea purpurea*, has shown significant increases in biomass and secondary metabolites under greenhouse conditions. In particular, the increase in CO₂ improved primary productivity and, in some cases, the concentration of bioactive compounds with pharmacological applications. These results suggest that CO₂ fertilization could become an essential tool for optimizing the production of high-value medicinal plants in protected environments [81]. In aromatic plants from the Lamiaceae family, such as *Mentha spicata* and *Thymus vulgaris*, ultra-high CO₂ levels (3000–10,000 µmol mol⁻¹) promoted increases of 150–220% in fresh weight and a significant improvement in morphogenesis, including a higher number of leaves and shoots. These responses highlight the potential of CO₂ not only to increase yield but also to optimize the structural characteristics of plants with culinary and medicinal applications [82].

Similarly, in legumes, such as soybean (*Glycine max*), elevated CO₂ at 650 µmol mol⁻¹ significantly increased leaf and root biomass and partially mitigated the negative effects of ozone (120 nl L⁻¹). At high CO₂ concentrations, the roots showed an 88% increase, suggesting that this technique could be effective in counteracting the impact of atmospheric pollutants on crops essential for food security. [83]. In *Castanea sativa*, elevated CO₂ at 700 µmol mol⁻¹ initially promoted photosynthesis and reduced nighttime respiration, improving the carbon balance. However, this effect was transient, as, by the end of the growing season, physiological rates returned to control levels. This behavior highlights the importance of dynamic CO₂ enrichment management to maximize its impact on productivity [84]. In Andean potato (*Solanum curtilobum*), elevated CO₂ at 720 µmol mol⁻¹ increased the total dry biomass by 66% and yields by 85%, highlighting its ability to optimize productivity in high-altitude agricultural systems. These results emphasize the potential of CO₂ as a tool to mitigate climate-related limitations in essential food crops [85].

CO₂ enrichment in controlled chambers has proven to be an effective strategy for enhancing the growth of various ornamental and floral species. In a study using an elevated CO₂ concentration of 1000 µL L⁻¹, species like *Saintpaulia ionantha* (African violet), *Rosa* (roses), *Kalanchoe blossfeldiana* (Kalanchoe), *Chrysanthemum × morifolium* (chrysanthemum), *Helxine soleirolii* (Irish ivy), *Hedera helix* (common ivy), and *Nephrolepis exaltata* (sword fern) were evaluated. All ornamental species showed a positive response to the increased CO₂ concentration, reflected in higher dry weights and more vigorous growth. In particular, roses and *Saintpaulia ionantha* showed significant increases in dry weight under CO₂ enrichment conditions. However, the addition of nitrogen oxides (NO_x) at a concentration of 0.85 µL L⁻¹ negatively affected the development of these species. In roses, stem growth was reduced, while in *Saintpaulia* leaf size decreased and the time to flowering increased, accompanied by a reduction in the number of flowers and flower buds [86].

Trees, such as *Betula pubescens*, have also shown significant increases in biomass under elevated CO₂ concentrations between 560 and 700 $\mu\text{mol mol}^{-1}$, especially when combined with moderate temperatures of 20 °C. In greenhouse experiments, CO₂ enrichment improved the relative growth rate (RGR) by up to 10%, increasing plant height, stem diameter, and biomass partitioning toward branches and roots. However, ozone (62 nmol mol^{-1}) significantly reduced this positive effect, highlighting the sensitivity of trees to atmospheric pollutants [87].

In forage grasses, such as *Phleum pratense*, *Lolium perenne*, and *Festuca pratensis*, CO₂ enrichment at 740 $\mu\text{mol mol}^{-1}$ increased the total biomass by 30%, regardless of soil type. However, ozone (50 nmol mol^{-1}) reduced the dry weight by 18% (the effects of carbon dioxide concentrations on three grass species grown in a mixture in two soil types at different ozone concentrations or temperatures). In *Elymus athericus*, a C3 grass, elevated CO₂ increased biomass production by 67%, but this effect was reduced by 31% when exposed to high UV-B radiation. The combination of both factors resulted in a net decrease of 8% in biomass, suggesting that CO₂ can partially offset the negative effects of UV-B radiation. These results highlight the complexity of interactions between environmental factors in grass performance under controlled conditions [88].

3.3. Application of Modeling and Simulation Techniques to Analyze the Distribution of CO₂ and Its Impact on Crops Within Protected Environments

Modeling and simulation of crop growth or the microclimate of greenhouses and plant factories provide fundamental tools to predict and optimize agricultural performance in these production systems [89,90]. Mathematical and computational models allow for the analysis of how plants respond to different levels of CO₂, light, and temperature, thus facilitating decision making regarding the best management practices. The articles discussed in this section address the development and validation of simulation models that replicate real conditions, contributing to improved production efficiency and reducing uncertainty in controlled cultivation systems.

CO₂ control in greenhouses or controlled environments is a key component for maximizing crop yield and optimizing photosynthesis. A decrease in CO₂ levels in closed environments negatively affects plant growth, which has led to the development of advanced models based on artificial intelligence. Among the most prominent strategies are enhanced genetic algorithms and fuzzy neural networks, which enable the efficient self-regulation of CO₂ enrichment. These tools have been successfully applied in crops, such as tomato, lettuce, and pepper, showing increases in yield and reductions in energy consumption. For example, studies on lettuce have reported a 15% increase in profitability by integrating economic and agronomic prediction models [91,92].

A mathematical model was also developed to optimize CO₂ enrichment in greenhouses, highlighting its fundamental role in photosynthesis and the efficient production of crops. This model correlates critical environmental factors with CO₂ concentrations measured at different depths through regression analysis. Experimental results demonstrated the model's ability to predict CO₂ emissions, validating its applicability in real-time monitoring and control systems. This simulation tool can enhance the efficiency of CO₂ enrichment strategies, maximizing agricultural profitability and reducing environmental impacts, establishing itself as a comprehensive approach for the sustainable management of greenhouses [93].

In crops like tomato, on-site microclimate modeling has proven essential for understanding and predicting the behavior of variables, such as temperature, CO₂ balance, and water vapor. Models based on experimental measurements have achieved correlations above 95% between simulated and observed conditions, validating their use for the design of climate control systems. Additionally, optimization techniques, such as particle

swarm optimization (PSO) and support vector machines (SVMs), have achieved accuracies of 96% in predicting the photosynthetic rate of tomato at different development stages, significantly contributing to crop efficiency [94,95].

In tomato, studies have also been developed to analyze the interaction between CO₂ and soil moisture. Backpropagation neural network models were used to predict the photosynthetic rate of tomato plants at different developmental stages (seedling, flowering, and fruiting), combining four levels of CO₂ concentration (450, 700, 1000, and 1300 µmol/mol) with three levels of soil moisture (low, moderate, and high). Environmental data were collected using sensor nodes and a photosynthesis analyzer, serving as the basis to train and validate the models. The results indicated that the models achieved high accuracy in predicting the photosynthetic rate, with coefficients of determination (R^2) greater than 0.95 and minimal errors ($RMSE \leq 1.22 \mu\text{mol m}^2\cdot\text{s}$) in all growth stages. Additionally, the models effectively described the relationship between CO₂ concentration and photosynthesis under different soil moisture conditions, identifying specific CO₂ saturation points. This approach provides a solid foundation for quantitatively regulating CO₂ enrichment in tomato crops in greenhouses, optimizing growth and development based on environmental conditions [96].

On the other hand, a simulation analysis combined biological and physical models to evaluate the impact of CO₂ enrichment on tomato cultivation. The models predicted the distribution of photosynthates, fruit growth, and enrichment costs, considering factors like leaks and ventilation. The results indicated that in winter in the UK, with closed ventilation, a CO₂ concentration close to 1000 ppm maximized the economic margin. In summer, with open ventilation at 5%, the optimal concentration was 500 ppm. The model also allowed for the evaluation of alternative strategies, demonstrating its usefulness in optimizing the economic performance of the crop under different climatic and management conditions [97].

In terms of CO₂ distribution and absorption in tomato crops in controlled environments, computational fluid dynamics (CFD) studies have shown that closed vents provide a more uniform distribution of CO₂ (CV of 8.7%) compared to open vents (CV of 8.8%) (Figure 4). Additionally, simulations have shown that although photosynthetically active radiation (PAR) partially compensates for CO₂ absorption on sunny days, no significant differences were observed between variable climatic conditions, highlighting the importance of optimizing both ventilation and CO₂ distribution to maximize photosynthetic efficiency [98].

In the case of cucumber, a study used a dynamic simulation model to analyze energy and mass exchanges in open and confined greenhouses, considering factors like ventilation, variable shading, cooling, and CO₂ enrichment. The model allowed for the evaluation of the distribution of critical parameters, including air and leaf temperatures, photosynthetically active radiation, and the dynamics of photosynthesis and transpiration. The results showed that heat pump systems, combined with variable shading, optimize cooling requirements, improving the photosynthetic environment in both open and confined greenhouses. This highlights the capability of modeling to predict and enhance the efficiency of climate control systems in agricultural production under protected conditions [99].

Strawberry cultivation has also benefited from innovative CO₂ enrichment strategies, analyzed through CFD modeling. Systems like the crop-localized CO₂ enrichment (CLC) method have been examined, with results showing that the use of these strategies increases CO₂ concentration by 264 µmol mol⁻¹, boosting the photosynthetic rate by 1.48 µmol m⁻² s⁻¹ and improving energy efficiency by 440%. As a result, total fuel consumption is reduced by 27% compared to full-enrichment systems. This approach

stands out for its feasibility in reducing operational costs and improving sustainability in protected cultivation systems [100].

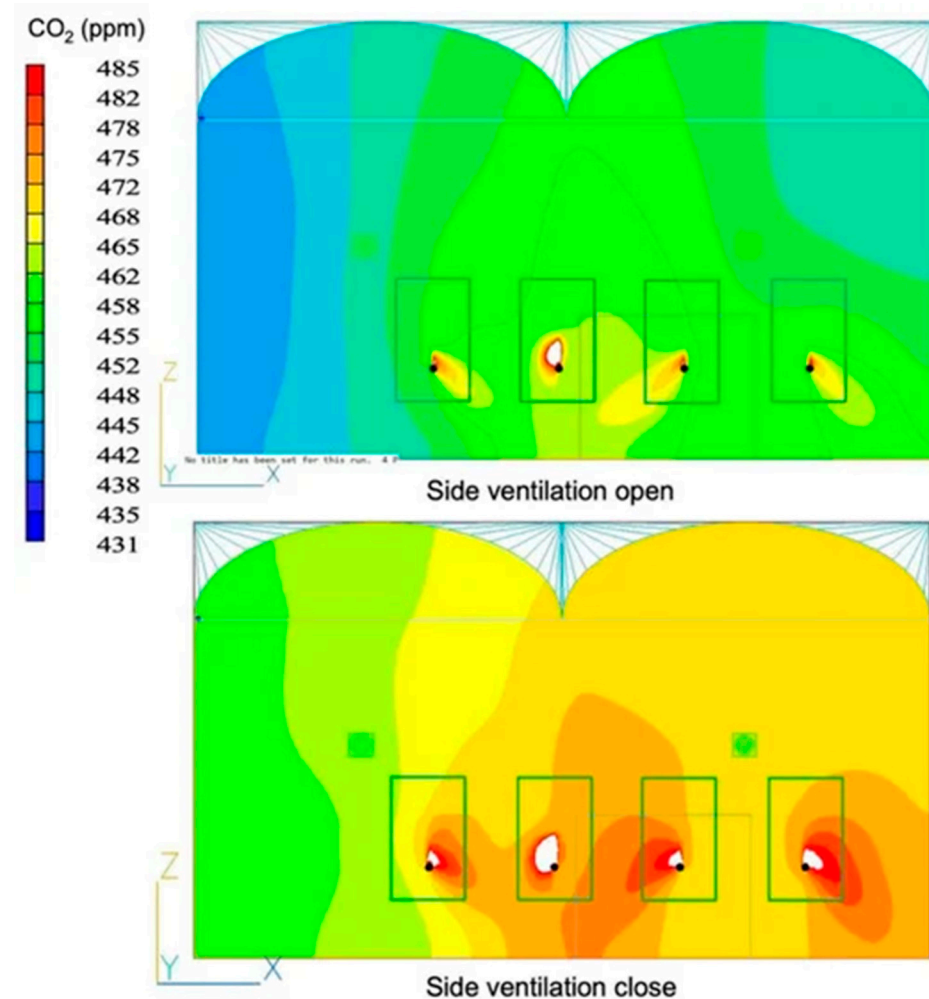


Figure 4. Variation in CO₂ concentration due to natural ventilation. Figure taken from Nuralisa et al. [98].

In strawberry, another study used physical, physiological, and biochemical models to evaluate the spatiotemporal variability of photosynthesis in a greenhouse with strawberry plants, linking the microclimates generated by environmental controls, such as roof ventilation and CO₂ enrichment (Figure 5). The results showed that photosynthetic distributions were highly variable and non-uniform, primarily influenced by air temperature and leaf boundary layer conductance, which were related to the energy budget and physiological properties of the leaves. CO₂ enrichment intensified this lack of uniformity through large variations in photosynthetic rates limited by Rubisco and RuBP. The spatial uniformity of photosynthesis fluctuated between 15% and 69% during the day. These findings emphasize the need to address this variability in future environmental control designs to optimize production in greenhouses [101].

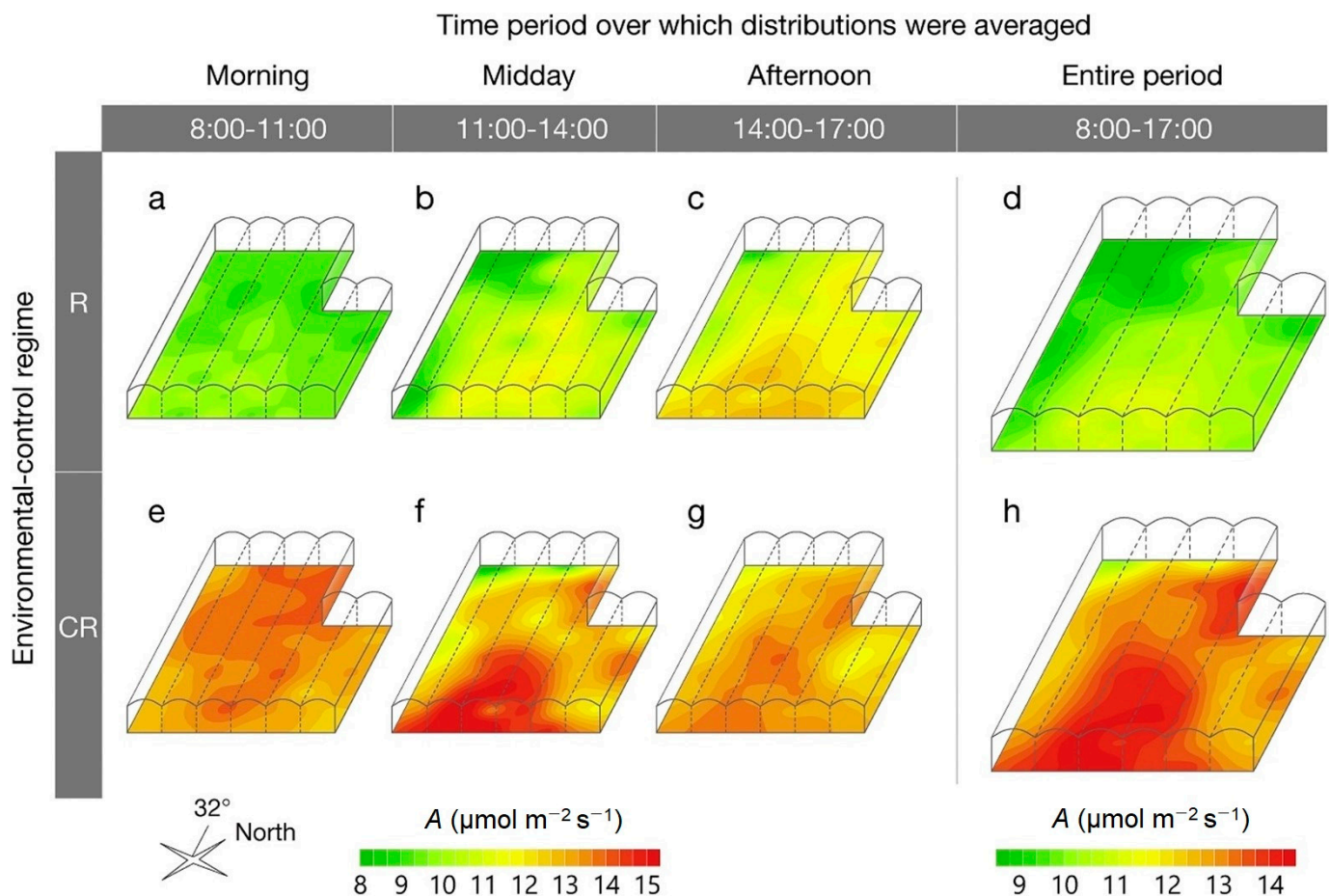


Figure 5. Spatial distributions of leaf photosynthetic rate A in the greenhouse, averaged, under two environmental control regimes: roof ventilation (R: day of the year 55 in 2018) and CO₂ enrichment in addition to roof ventilation (CR: day of the year 57 in 2018). Figure taken from Kimura et al. [101].

On the other hand, in a greenhouse dedicated to mango cultivation, a short-term memory model (LSTM) was used to predict CO₂ concentration, a key factor for optimizing photosynthesis and crop production. The model used microclimatic data, such as temperature, relative humidity, solar radiation, and CO₂ concentration, collected through nine sensors distributed throughout the greenhouse. Using historical data measured every 10 min over a period of 16 months, the LSTM achieved accurate CO₂ concentration predictions ($R^2 = 0.78$) up to 2 h ahead. This approach highlights the potential of modeling and simulation using artificial intelligence to manage CO₂ enrichment, improving photosynthetic efficiency in greenhouses [102].

Additionally, the impact of CO₂ enrichment has also been analyzed in ornamental crops, such as *Scaevola aemula* (Fairy Fan Flower). The use of coupled models in this plant allowed for high-precision predictions of the effects of CO₂ and supplemental lighting on photosynthesis and transpiration, highlighting the usefulness of these models for optimizing environmental management strategies. Among the relevant results, it was reported that net CO₂ assimilation showed saturation at intercellular CO₂ concentrations above 600 $\mu\text{mol mol}^{-1}$, with a maximum rate of 23.1 $\mu\text{mol m}^{-2} \text{s}^{-1}$ at 25 °C and PAR of 1500 $\mu\text{mol m}^{-2} \text{s}^{-1}$. Assimilation rates decreased significantly at leaf temperatures below 20 °C. Finally, the simulations demonstrated that the model is useful for exploring the effects of supplemental lighting and CO₂ enrichment on canopy photosynthesis and transpiration, positioning it as a promising tool for optimizing environmental conditions in the production of this plant [103].

In hydroponic crops, specifically in *Capsicum annuum* L. (bell pepper), the use of multivariable models, such as the FvCB (Farquhar, von Caemmerer, and Berry) model, has achieved a precision of $R^2 = 0.91$ and low RMSE errors. Additionally, a linear relationship was found between the total nitrogen content in the leaves, which increased with vertical position, and the physiological parameters of the FvCB model, such as the maximum carboxylation capacity and the maximum electron transport rate. These results highlight that the FvCB model is more suitable for linking photosynthetic responses with nitrogen content and for establishing environmental management strategies and CO₂ enrichment in greenhouses to optimize photosynthesis and crop growth [104].

In another study, a Functional Structural Plant Model (FSPM) was used to analyze the effects of CO₂ enrichment at 900 $\mu\text{mol mol}^{-1}$ on photosynthesis, morphology, and yield in sweet pepper. Growth, yield, and photosynthesis were regularly measured, incorporating 3D phenotyping and gas exchange data into the model. The results showed that the impact of CO₂ varied depending on the growth stage. In the early stages, enrichment accelerated the growth transition and increased the yield by redirecting photosynthates toward the fruits. However, in later stages, efficiency decreased due to limited leaf area expansion caused by excessive partitioning of photosynthates. This approach highlighted how balancing the source–sink relationship of photosynthates through proper agricultural practices could maximize productivity in CO₂ enrichment systems [105].

Finally, in solar greenhouses, which are a promising alternative for CO₂ capture and reuse, mathematical models have been implemented to predict CO₂ captures of up to 140 g CO₂/m² per day under high planting density conditions, improving photosynthesis and reducing water consumption [106]. Similarly, computational models have been designed to simulate the transient performance of greenhouses in hot and arid climates, with a focus on energy efficiency and continuous CO₂ enrichment. The model divides the greenhouse into interactive components—the cover, soil, growing medium, air space, and crop—while integrating a physiological tomato model that responds to photosynthetically active radiation, leaf temperature, and CO₂ concentration. To minimize the use of conventional energy, technologies like solar air heaters, thermal rock accumulators, and movable thermal screens were incorporated. Additionally, systems like a total enthalpy wheel and an evaporative cooler were simulated to dehumidify and cool without environmental ventilation, allowing for continuous CO₂ enrichment. The results validate the model's ability to optimize the design of sustainable greenhouses and enhance crop performance in protected environments [107].

3.4. CO₂ Injection and Enrichment Technologies

CO₂ injection technologies in greenhouses represent a key innovation to maximize agricultural productivity and improve crop quality. The studies in this section focus on the design, development, and implementation of CO₂ injection systems that enable homogeneous and efficient distribution within the cultivation environment. The integration of injection technologies with ventilation, heating, and cooling systems has proven to be an effective tool for increasing photosynthesis and reducing operational costs, thus contributing to the advancement of precision agriculture systems [16].

CO₂ enrichment in plant factories and greenhouses can be achieved through various sources, each with its own characteristics, advantages, and limitations, significantly improving the performance of crops like celery and lettuce. Compost, generated from crop residues and animal manure, represents an economical CO₂ source for enriching greenhouses, while also contributing to the reduction of agricultural carbon, nitrogen, and phosphorus emissions. This practice can produce CO₂ levels ranging from 1000 to 1500 ppm, along with gaseous coproducts, such as ammonia (NH₃). Its main advantage lies

in reducing waste by providing an optimal environment for fermentation and eliminating composting byproducts. However, it requires proper environmental control to maintain ideal composting conditions and avoid contamination. The secondary products of compost can also be reused, although this process requires an appropriate environment for secondary composting. Additionally, compost-based enrichment has limitations related to uneven CO₂ distribution and the inability to precisely control it in automated greenhouses. Therefore, the integration of controlled storage and release technologies could optimize its use in less automated environments [108–110].

Similarly, CO₂ generation through chemical reactions, such as those involving sodium bicarbonate and acids or thermal decomposition, is a relatively economical and efficient technique for obtaining pure CO₂ in a quantitative manner [14]. Although, in theory, the CO₂ production rate can be controlled, in practice, the process presents operational challenges that may result in uncontrolled CO₂ release, wasting resources and potentially harming the crops [111]. In some cases, ammonium bicarbonate is used as the raw material, generating byproducts that can be utilized as fertilizers. However, this option involves risks of ammonia gas poisoning, making it essential to implement NH₃ filtration systems to ensure safety [112]. While it offers advantages, this technology requires careful management to be considered a controllable source.

Natural gas represents a controlled source of CO₂ through the combustion of fossil fuels in boilers or cogeneration systems. It is a common CO₂ source with concentrations up to 1500 ppm, which is particularly useful on cold days to heat the greenhouse. It is more economical than pure CO₂, but it has significant limitations, such as reliance on fluctuating fossil fuel prices, negative environmental impact, and the need for emission purification before use. Additionally, it is not suitable for warm regions [15,113]. A study evaluating the integration of heating and CO₂ enrichment with advanced climate control, where the use of activated carbon for CO₂ capture through adsorption was highlighted, reported that the system achieved a storage capacity of 46.7 g kg^{−1} and 99.99% removal of contaminants. This ensures optimal CO₂ levels at 851.0 ± 262.8 mg Nm^{−3} in the greenhouse, with an average enrichment time of 2.18 ± 0.92 h day^{−1}. During a pilot evaluation in winter (November to February), the system demonstrated stability and an 18% increase in tomato crop yield compared to a control [114].

Similarly, a CO₂ clathrate-based air conditioning system for greenhouses has been evaluated, which is designed to combine CO₂ enrichment and cooling under high-temperature conditions. This approach aims to optimize photosynthesis in crops while reducing operational costs associated with traditional cooling. Although the clathrate's capacity proved sufficient to maintain stable CO₂ levels, promoting photosynthesis in tomato plants, the system still does not fully replace electric cooling systems. However, combining this method with innovative cooling approaches could overcome these limitations, offering a promising solution for regions with hot climates [115]. This advancement highlights the importance of integrating emerging technologies to achieve efficient and sustainable CO₂ enrichment in greenhouses, complementing traditional systems, and optimizing agricultural productivity.

Another alternative is biogas, obtained from landfills or through anaerobic digestion of dairy manure, which can reach CO₂ levels of up to 2500 ppm. This method is environmentally friendly and promotes the conversion of waste into energy. However, it requires additional renewable energy production systems, as well as the purification of emissions that contain compounds like SO₂, H₂S, and other volatile organic compounds. This approach is ideal for promoting carbon neutrality, although it presents logistical and technical challenges [116,117]. On the other hand, wood pellets, almond shells, pine wood, and olive pits are alternative sources of CO₂ that also generate byproducts, such as CO and NO_x. These options are more sustainable, as they reduce reliance on fossil fuels and

can simultaneously heat and enrich greenhouses. However, they require additional equipment, such as precipitators, filters, and catalytic converters, to ensure safe and efficient operation [118,119].

Ambient air is also a simple and low-cost option for CO₂ supply, allowing for concentrations of approximately 350 ppm (currently exceeding 420 ppm). Although easy to implement, its effectiveness is limited, particularly during winter, when CO₂ concentrations may be insufficient to meet plant requirements [68,120]. According to Stanghellini et al. [121], on mild days, ventilation is sufficient to compensate for CO₂ depletion, but, on cold days, heating costs may outweigh the benefits of passive enrichment. Therefore, ventilation should be considered a complementary strategy in warm or moderate climates. This source provides a safe and pure option for carbon enrichment, allowing any desired concentration to be achieved [122]. Its main advantage is flexibility, as it can be used at any time without the need for additional preparation. However, the relatively high cost of pure CO₂ limits its widespread adoption [123,124].

Pure liquefied CO₂ stands out as a safe and highly efficient source of enrichment in modern greenhouses and plant factories typically injected from storage tanks through controlled valves. This method is ideal for modern greenhouses due to its key characteristics; it is pure, safe, and portable, enabling efficient enrichment [125]. Research studies, such as those by Chalabi et al. [113] and Kuroyanagi et al. [124], have highlighted the economic feasibility of this system, especially when combined with optimal control strategies. However, its high initial cost limits its implementation in small or medium-sized greenhouses. These options represent diverse alternatives that, depending on the specific cultivation conditions and technical and economic constraints, can significantly contribute to CO₂ enrichment in greenhouses and plant factories.

Other emerging technologies, such as carbon capture, utilization, and storage, offer innovative solutions by employing adsorption processes to capture CO₂ directly from the atmosphere, mitigating climate impact and providing a continuous supply for greenhouses [126,127]. Devices like the rotating packed bed (RPB) stand out in this context due to their ability to operate continuously with lower energy costs and high efficiency in heat and mass transfer. Numerical models and advanced simulations have identified optimal adsorbent materials and ideal operating conditions, laying the foundation for optimizing these systems in modern agricultural applications [128]. These technologies represent a significant advancement toward the sustainable and efficient management of CO₂ in greenhouses.

According to the above, a three-dimensional thermal and mass imbalance model was developed to analyze heat and mass transfer characteristics in bisectional RAW systems using different adsorbent materials, such as activated carbon, zeolite 13X, and Mg-MOF-74 (Figure 6). The results indicated that zeolite 13X and Mg-MOF-74 exhibit significantly superior performance compared to activated carbon, with minimal differences between them due to temperature fluctuations in the adsorption and desorption sectors. However, factors like rotation speed and airflow rate negatively impact CO₂ enrichment performance, reducing efficiency when the overall mass transfer coefficient is lower than 0.002 s⁻¹ [129,130]. Despite the promising simulation results, this technology requires experimental validation in large-scale greenhouses or plant factories to assess its operational feasibility and performance under real conditions. This step is crucial to confirm the RAW system's ability to optimize crop growth by maintaining optimal CO₂ concentrations in a sustainable and efficient manner.

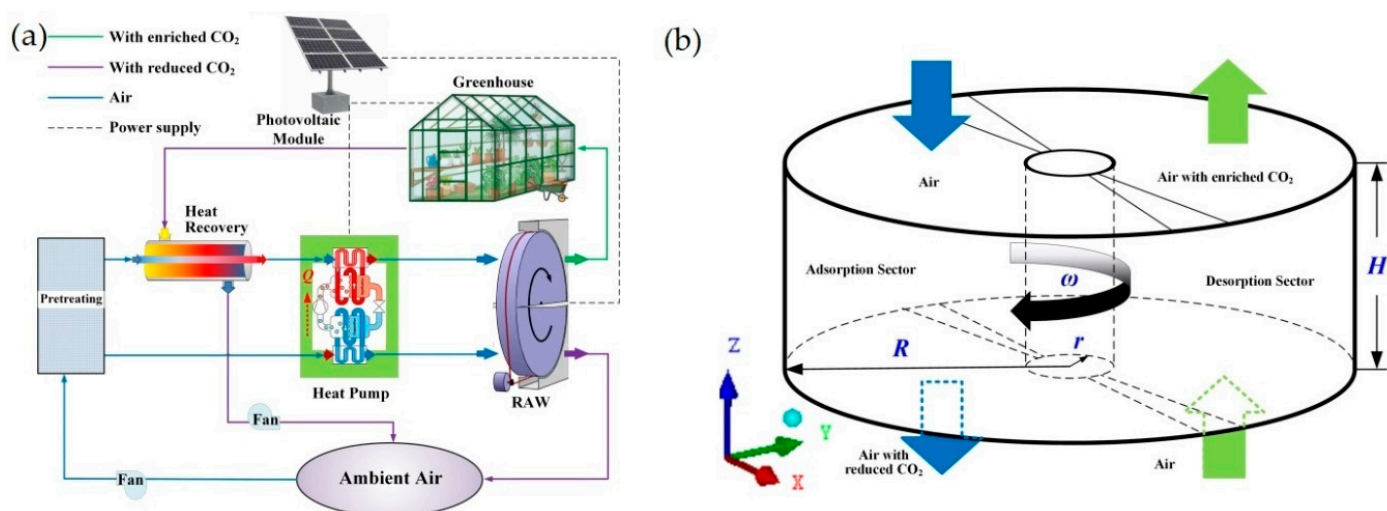


Figure 6. (a) Schematic of the CO₂ enrichment system for a greenhouse using a rotating adsorption wheel. (b) Geometry of a bi-sectional regenerative rotating adsorption wheel. Figure adapted from Tang et al. [129].

A relevant aspect in protected agriculture and CO₂-enriched environments is the quantification of CO₂ use efficiency (CUE), defined as the ratio between the net photosynthetic rate and the CO₂ supply. In greenhouses, CUE is typically below 60%, influenced by factors like leakage, excessive supply, and environmental conditions [124,125,131]. To improve this efficiency, strategies have been developed focusing on spatial distribution, period configuration, and concentration control. Uniform CO₂ distribution is essential to maximize its utilization. Concentration is generally lower in the canopy with high leaf density, where demand is highest [73,100]. Solutions, such as perforated pipes and internal air circulation devices, have proven effective in improving distribution and reducing waste, while also enhancing yield in crops like strawberries [73,132]. Regarding the enrichment period, CO₂ supply in the morning can be as effective as continuous supply, promoting biomass accumulation and fruit quality without significantly increasing energy consumption [62,133]. However, intermittent strategies have shown mixed results depending on species and cultivation conditions, while the effects of nocturnal enrichment remain uncertain and species-specific [134].

The control of internal CO₂ concentration relative to the external environment is key to maximizing CUE. Moderate concentrations of 550–650 $\mu\text{mol mol}^{-1}$ have been shown to be sustainable and cost-effective, enhancing yield with a lower environmental impact [135]. However, higher concentrations can alter the nutritional composition of the product, increasing sugars, such as glucose and fructose, while reducing essential amino acids and minerals [136]. The optimal concentration should be adjusted to specific production objectives while also considering other environmental factors to minimize costs and maximize benefits. In large-scale or multi-level greenhouses, wireless sensor networks (WSNs) have revolutionized environmental monitoring by replacing traditional wired systems, which were costly and limited in flexibility. These networks integrate optical sensors, such as infrared and fiber optic sensors, and electrochemical sensors, including metal oxide and polymer-based sensors, to accurately measure CO₂ concentrations [16,137,138]. Communication technologies, such as Bluetooth, ZigBee, and 5G networks, facilitate data transmission to remote control platforms, enabling detailed analysis and informed decision making to optimize crop growth [139]. Additionally, advanced protocols ensure data reliability in scenarios with mobile sensors and complex CO₂ distributions [140,141].

Dynamic control of CO₂ enrichment requires consideration of factors like photosynthesis, respiration, and ventilation. These processes, along with climatic variations, such

as wind speed and solar radiation, generate daily CO₂ concentration fluctuations in a “U” shape, which must be managed through adaptive strategies [142]. Advanced control methods, such as optimal control, integrate dynamic predictions and environmental parameter optimization, enhancing economic efficiency and facilitating the development of precision greenhouses [143].

3.5. Plant Factory with Artificial Lighting (PFAL) and CO₂ Enrichment

Plant Factories With Artificial Lighting (PFAL) represent a revolution in agricultural production by utilizing closed systems that optimize plant growth under artificial lighting and CO₂ enrichment [144]. The studies in this section examine how elevated CO₂, combined with artificial lighting and controlled microclimatic conditions, enhances the yield of high-demand crops. PFAL not only maximizes production per unit area but also enables vertical farming, making it a viable solution for intensive production in urban environments and space-constrained areas [145].

The development of closed Plant Factories With Artificial Lighting and advanced environmental control has optimized agricultural production through the precise management of factors like light, temperature, and CO₂. These facilities are distinguished by their high resource use efficiency, and they are particularly effective for leafy vegetable crops, such as lettuce, cabbage, spinach, and kale. Experimental studies have demonstrated that CO₂ concentrations of up to 1000 ppm and temperatures of 25 °C promote accelerated growth and increased dry matter accumulation in lettuce and cabbage crops, while light intensities up to 480 $\mu\text{mol m}^{-2} \text{s}^{-1}$ do not reach saturation levels [146].

In PFAL systems, the effects of different air velocities (0.1 to 0.8 $\text{m}\cdot\text{s}^{-1}$) on photosynthesis in tomato seedling canopies have also been investigated within closed production modules under CO₂ concentrations of 0.4 and 0.8 $\text{mmol}\cdot\text{mol}^{-1}$. The experiment, conducted in a wind tunnel chamber, evaluated the net photosynthetic rate by measuring the difference in CO₂ concentrations between the system’s inlet and outlet. Environmental conditions included a photosynthetic photon flux (PPF) of 0.25 $\text{mmol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, a temperature of 23 °C, and a relative humidity of 55%. The results showed that the net photosynthetic rate of the tomato canopy increased with higher air velocities, saturating at 0.2 $\text{m}\cdot\text{s}^{-1}$. At a velocity of 0.4 $\text{m}\cdot\text{s}^{-1}$, the photosynthetic rate was 1.3 times higher than at 0.1 $\text{m}\cdot\text{s}^{-1}$. Additionally, an elevated CO₂ concentration of 0.8 $\text{mmol}\cdot\text{mol}^{-1}$ increased the photosynthetic rate by 20% compared to the standard concentration of 0.4 $\text{mmol}\cdot\text{mol}^{-1}$. These findings highlight the importance of controlling air circulation to maximize canopy photosynthesis under both elevated and normal CO₂ conditions in closed plant production modules [147].

Recently, air flow control has continued to be analyzed, as it is crucial for regulating the canopy microclimate in plant factories (PFAL). Therefore, a CFD simulation model was developed and validated, allowing for the prediction of key parameter distributions, such as radiation, leaf temperature, and air velocity, within a soybean canopy. The results showed that at low air velocities (<0.57 $\text{m}\cdot\text{s}^{-1}$), the microclimate was primarily influenced by lamp radiation, while increasing inlet velocity improved microclimate uniformity. These findings complement previous studies by highlighting how the interaction between airflow and CO₂ concentration in closed modules optimizes plant growth and development in controlled systems [145].

In terms of operational efficiency, technologies, such as mini air handling units (MAHUs), have proven to be key in the continuous regulation of CO₂ levels in closed systems. In plant factories dedicated to lettuce production, these units enable the maintenance of stable CO₂ concentrations, optimizing energy efficiency and reducing the reliance on external inputs [148]. Additionally, life cycle assessment (LCA) of plant factories has shown that although they require higher energy inputs due to artificial lighting, they

achieve greater efficiency in the use of water, pesticides, and land compared to traditional greenhouses. Notably, CO₂ emissions per kilogram of crop are significantly lower in hybrid systems that combine natural and artificial lighting [149].

CO₂ enrichment not only enhances yield but also improves the nutritional quality of crops. In red leaf lettuce, CO₂ concentrations of 1000 ppm increased both yield and the accumulation of bioactive compounds, such as flavonoid glycosides and caffeic acid derivatives [150]. Similarly, in kale, CO₂ concentrations between 700 and 1600 ppm promoted physical growth and the accumulation of glucosinolates. However, higher temperature and relative humidity levels reduced these bioactive compounds, highlighting the importance of precise environmental management [151].

In a previous study, environmental factors were evaluated to optimize the growth of *Lactuca sativa* var. *crispa* cv. Grand Rapids. CO₂ enrichment and a light intensity of approximately 200 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ resulted in maximum photosynthetic and growth rates, particularly at CO₂ concentrations of 1000 and 2000 $\text{mg}\cdot\text{L}^{-1}$. However, nutrient levels did not significantly influence growth. These results highlight the importance of adjusting CO₂ concentration and light intensity to maximize productivity in controlled cultivation systems [152]. On the other hand, in turnips, light intensities of 316 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ and 24 h photoperiods promoted accelerated tuber growth. Meanwhile, in lettuce, stomatal resistance and the CO₂ compensation point increased with higher CO₂ concentrations and light intensity, once again highlighting the physiological differences between species [153].

In terms of photosynthetic efficiency, research on lettuce under CO₂ concentrations of 800 and 1600 ppm showed a 41% increase in light use efficiency (LUE), with yield improvements of up to 44%. These findings reinforce the role of CO₂ in optimizing resource use in closed cultivation systems [10]. In *Amara* mustard crops, extremely high CO₂ concentrations of 2800 ppm significantly increased the biomass. However, under far-red light, a reduction in leaf area was observed, highlighting the variability in physiological responses among crops [154].

The benefits of CO₂ enrichment also extend to seedling propagation. Photoautotrophic micropropagation, which eliminates the need for exogenous organic compounds in the culture medium, has proven to be an efficient technique for producing high-quality, genetically identical seedlings. This method, when combined with precise CO₂ and light control, enhances seedling physiology and reduces operational costs, making it ideal for both woody and herbaceous species [155].

The design of plant factories has evolved to incorporate integrated strategies that maximize photosynthesis and gas exchange. In lettuce, three-way interactions between CO₂ concentration, light intensity, and air velocity increased net photosynthesis by 51%, while fresh and dry weight improved by 36% and 20%, respectively (Figure 7) [156]. Similarly, canopy photosynthetic models, such as those developed for romaine lettuce, have enabled precise predictions of CO₂ requirements throughout the crop cycle, allowing for the efficient adjustment of concentrations based on the growth stage [157].

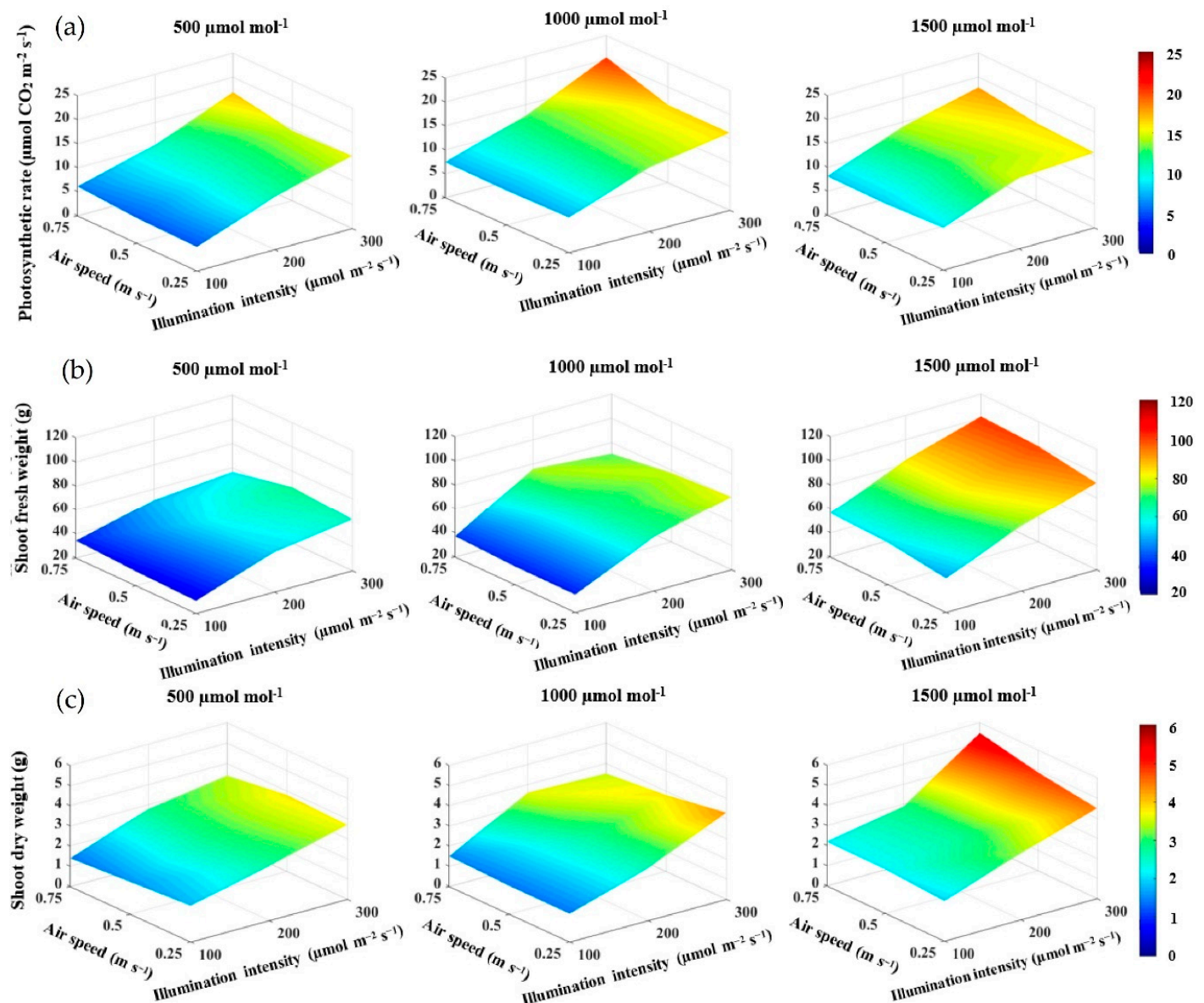


Figure 7. Effect of CO₂ concentration, light intensity, and air velocity on (a) photosynthetic rate; (b) fresh weight; and (c) dry weight of lettuce plant shoots grown under artificial lighting. Figure adapted from Ahmed et al. [156].

3.6. Economic Aspects and Energy Efficiency of CO₂ Enrichment in Protected Agriculture

The implementation of CO₂ enrichment technologies must be assessed from an economic perspective to ensure long-term feasibility and profitability. Analysis of various studies highlights advancements in energy efficiency and the economic aspects related to CO₂ enrichment in protected cultivation systems and plant factories. The studies provide key insights for optimizing energy and resource use, with a focus on economically significant crops.

CO₂ enrichment in protected systems, such as greenhouses and plant factories, is crucial for enhancing photosynthesis and crop yields in species like tomato, cucumber, sweet pepper, white clover, and lettuce. However, the high energy consumption associated with these systems remains a significant challenge. In greenhouses located in arid regions, for example, CO₂ enrichment combined with advanced climate control technologies significantly improved light use efficiency (LUE) and crop yield, demonstrating a positive correlation between greenhouse technological level and productivity [158]. The ISBA-A-gs scheme

(Soil–Biosphere–Atmosphere interactions responsive to CO₂), applied to perennial ryegrass pastures, enabled modeling the effects of CO₂ enrichment in combination with nitrogen fertilization. The results showed that this approach can increase biomass and the leaf area index, providing a valuable quantitative framework for optimizing the management of crops like white clover and similar species [159].

In commercial strawberry greenhouses, CFD simulations demonstrated that CO₂ concentrations near 500 $\mu\text{mol}\cdot\text{mol}^{-1}$ maximize energy efficiency and photosynthetic performance. This result highlights the importance of defining optimal enrichment concentrations to ensure the economic and environmental sustainability of strawberry production [13]. In rooftop greenhouses integrated with buildings, the implementation of predictive control strategies based on nonlinear models (NMPCs) optimized the use of excess heat and air from the building for crops like tomato and lettuce. This system achieved an average energy savings of 15.2%, reducing operational costs and CO₂ emissions, positioning itself as an efficient solution for urban agriculture [160].

The use of CFD models in semi-closed greenhouses with tomato allowed for the simulation and validation of CO₂ distribution, demonstrating that controlled enrichment systems enhance distributed photosynthesis and reduce losses due to ventilation. This approach is particularly valuable for optimizing tomato yield in semi-closed systems [11]. In plant factories, crops like lettuce demonstrated efficient resource use when combined with energy-saving strategies, such as efficient lighting and air economizers. Water and energy consumption varied depending on geographical location, emphasizing the importance of establishing these facilities in regions with cleaner power grids to reduce associated carbon emissions [161].

In tomato greenhouses, optimal CO₂ enrichment control based on economic models demonstrated that profitability increases significantly by continuously adjusting CO₂ concentrations and utilizing heat generated through cogeneration. This approach optimized the financial margin between the cost of natural gas, CO₂, and crop value [15]. In studies on white clover and UV-B radiation, results indicated that reduced radiation levels could negatively impact crop growth and physiology, highlighting the importance of considering greenhouse characteristics to maximize the benefits of CO₂ enrichment in this and other crops [162].

CO₂ enrichment, combined with advanced technologies, such as CFD, NMPC, and cogeneration systems, presents a significant opportunity to enhance the sustainability and profitability of protected crop production. Crops, such as tomatoes, strawberries, lettuce, cucumbers, sweet peppers, and white clover, directly benefit from these innovations, which optimize resource use and reduce environmental impact. However, implementation must be tailored to the specific crop and location conditions, balancing energy efficiency, water consumption, and carbon emissions to achieve a sustainable and economically viable production system.

3.7. Environmental Aspects and Sustainability of CO₂ Enrichment in Protected Agriculture

CO₂ enrichment in protected agriculture also has environmental implications, as it can contribute to reducing carbon emissions and fostering more sustainable agricultural systems. The articles in this section explore the role of CO₂ in mitigating the carbon footprint, as well as the synergies between CO₂ enrichment and resource conservation, including water and energy. Additionally, sustainable management strategies are discussed, aiming to balance agricultural growth with environmental preservation, promoting an agricultural approach that addresses the challenges of climate change.

Plant factories and greenhouses are fundamental pillars of protected agriculture, enabling controlled and climate-resilient production. However, their environmental im-

pact varies significantly depending on the technology employed [163]. While traditional greenhouses rely on sunlight and nitrogen fertilizers, leading to significant emissions of nitrous oxide (N₂O) [164], plant factories rely on artificial lighting, which increases energy consumption and associated methane (CH₄) emissions, particularly in countries with energy grids based on fossil fuels [165]. These short-lived greenhouse gases, which have contributed significantly to global warming over the past decade [166], underscore the importance of assessing the climate impact of these technologies from a holistic perspective.

In a global context, the transition to decarbonized power grids is key to reducing the climate burden associated with protected agriculture. Countries like China, the world's largest vegetable producer [167], have implemented policies to reduce coal use in electricity generation, which could significantly enhance the climate sustainability of plant factories [168]. However, traditional assessment methods based on Global Warming Potential (GWP) do not adequately capture the dynamic and cumulative contributions of gases, such as methane and nitrous oxide. In this regard, the Technological Warming Potential (TWP) model provides a more robust tool for comparing agricultural technologies by considering their comprehensive impacts [169].

This approach has enabled a global comparison of the climate impact of crop production, such as lettuce, in Plant Factories With Artificial Lighting versus traditional greenhouses. While plant factories can enhance climate resilience in urban vegetable production, their carbon footprint is significantly higher than that of traditional greenhouses due to the high energy consumption associated with carbon-intensive sources like coal. An analysis showed that the Global Warming Potential (GWP) of PFALs (14.9 kg CO₂-eq·kg⁻¹) exceeds that of greenhouses (0.27 kg CO₂-eq·kg⁻¹) by more than 50 times. Decarbonizing the power grid could reduce plant factory emissions, lowering methane contributions to 4% by 2050. However, they would still not surpass greenhouses in climate sustainability unless predominantly powered by hydroelectric energy. This analysis underscores the need to advance toward protected agricultural systems that are not only highly productive but also globally sustainable, adapting to the energy and climate demands of each region [166]. These same conclusions were reported in a study conducted in five U.S. cities on lettuce cultivation [170].

CO₂ enrichment in greenhouses is emerging as a key strategy for mitigating the carbon footprint and optimizing water and energy resource use. Implementing CO₂ at the air intake through membrane materials has been shown to significantly reduce evaporation rates in greenhouses, leading to lower water loss—one of the most critical inputs in global agriculture [171]. On the other hand, combining lower temperatures with CO₂ enrichment in well-sealed greenhouses for lettuce production has reduced fuel consumption by 7% during critical periods, achieving a balance between productivity and economic sustainability [172]. In loblolly pine (*Pinus taeda*) seedlings, CO₂ enrichment at 700 ppm significantly increased growth even under water stress conditions. These results suggest that elevated CO₂ can mitigate the negative effects of water stress, providing substantial benefits to woody crops while balancing competition with weeds [173].

In semi-closed systems, CO₂ enrichment can compensate for light reduction caused by ventilation and cooling technologies. In tomato crops, increasing CO₂ concentrations from 400 to 1000 µmol·mol⁻¹ enhanced net photosynthesis by 51%, while transpiration decreased by up to 8%. This improved photosynthetic water use efficiency by 60%, highlighting the potential of these practices to enhance sustainability in agricultural production [27]. Solar greenhouses stand out as promising infrastructures for the capture and biological utilization of CO₂. Mathematical models suggest that these structures can enhance CO₂ capture by up to 1 g·m⁻²·day⁻¹, provided that alternative microclimate control methods are adopted. In

addition to carbon capture, these practices can increase crop productivity and reduce water consumption [106].

In tomato crops, CO₂ enrichment through cylinders or combustion showed that only 60% of the supplemented CO₂ was effectively utilized by the plants due to ventilation losses. However, the application of precise methods to measure fossil-derived carbon has enabled the optimization of enrichment strategies to maximize efficiency, promoting better integration of supplemental CO₂ into crop physiology [174]. Similarly, for this crop, the implementation of a neural-network-based model allowed for the prediction of daily photosynthesis dynamics under varying CO₂ concentrations, temperature, and light conditions. This model identified optimal CO₂ enrichment levels, increasing net photosynthesis by up to 40%. This approach underscores the importance of modeling and simulation in the precise regulation of gaseous fertilizers, providing a robust theoretical foundation for sustainable protected agriculture [175].

CO₂ enrichment can also coincide with environmental challenges, such as salinity, which results from frequent irrigation and intensive fertilization. A study evaluated the effects of exogenous spermidine (Spd, 0.25 mM), a polyamine, under combined conditions of elevated CO₂ (800 ppm) and isosmotic salt stress [150 mmol·L⁻¹ NaCl and 100 mmol·L⁻¹ Ca(NO₃)₂] in tomato crops. The results showed that salt stress significantly reduced parameters like plant fresh and dry weight, water content, and root characteristics. However, both elevated CO₂ and Spd mitigated these reductions, with their combined application being the most effective in enhancing salt stress tolerance. This effect was associated with improved osmotic adjustment and enhanced antioxidant capacity, reducing oxidative stress [176].

In a study on soilless greenhouse cultivation, models were developed to calculate CO₂ supply loads and size using gaseous fertilization equipment, considering sources like natural gas boilers and liquefied CO₂. This approach enables the design of systems that optimize carbon use and respond to the specific needs of each greenhouse, promoting more efficient agricultural infrastructure design [176].

The combination of bioenergy with carbon capture, storage, or utilization presents a sustainable solution to the Energy–Water–Food Nexus, addressing the challenge of meeting food demand while adapting to climate change. This approach integrates energy and agricultural systems by using CO₂ captured from a biomass-based integrated gasification combined cycle to enrich agricultural greenhouses. The results highlight synergies that enhance crop productivity by 13.8% and reduce water requirements by 28%, achieving negative CO₂ emissions and a levelized cost of USD 0.35 per kilogram of agricultural product. This model demonstrates significant potential for food and climate sustainability through the integration of energy and agricultural systems [177].

Rubisco, a key enzyme in the Calvin cycle, plays a crucial role in carbon sequestration by fixing atmospheric CO₂ during photosynthesis. Under current conditions of 400 ppm CO₂, photosynthesis in C3 plants, such as Italian lettuce, milk Chinese cabbage, and Shanghai green onion, is limited. However, increasing the CO₂ concentration to 1000 ppm in plant factories significantly enhances Rubisco activity, boosting growth and yield in these crops. With average carbon fixation rates of 1.2 g·m⁻²·h⁻¹, 2.5 g·m⁻²·h⁻¹, and 2.3 g·m⁻²·h⁻¹, respectively, a one-million-square-meter plant factory could capture approximately 10,000 tons of CO₂ annually [178]. This demonstrates that large-scale plant factories are not only an effective tool for increasing agricultural productivity but also a sustainable solution for mitigating carbon emissions.

It is also important to mention that carbon capture from biological sources through smart plant factories represents an innovative and sustainable approach to mitigating CO₂ emissions in industrial environments. These controlled infrastructures utilize exhaust gas

emissions to optimize the growth of *Pennisetum giganteum*, a species known for its high photosynthetic efficiency, rapid growth, and industrial value. A life cycle analysis (LCA) conducted in an automated plant factory demonstrated that 67% of the carbon emissions stem from lighting, while the growth of *Pennisetum giganteum* effectively mitigates these emissions, achieving a net reduction of 0.35 kg of CO₂ equivalent per kilogram produced. Therefore, a plant factory with dimensions of 3 × 6 × 2.8 m can reduce 174 kg of carbon emissions annually, achieving carbon sequestration that is 56% higher than that of open-field crops [179]. This model, in addition to offsetting carbon emissions in industrial parks, highlights the ability of plant factories to integrate environmental mitigation with agricultural production, establishing them as a key tool for sustainability and resilience in the face of climate change.

4. Challenges and Knowledge Gaps

CO₂ enrichment has positioned itself as one of the most promising strategies to improve growth and yield in protected agricultural systems, such as greenhouses and plant factories. However, the implementation of this technique faces various scientific, technological, and economic challenges. While significant improvements have been recorded in biomass, photosynthesis, and water use efficiency, numerous knowledge gaps remain regarding long-term effects, interactions with other environmental factors, and variations in responses across different species and crop varieties. Additionally, the economic viability and energy sustainability of CO₂ enrichment systems are areas of ongoing debate.

In this context, identifying areas where scientific evidence is limited is essential for advancing the efficient adoption of these technologies. Furthermore, it is crucial to set priorities for future research to address uncertainties and optimize CO₂ use in protected agricultural systems. This section provides a detailed analysis, first examining the main challenges and knowledge gaps in the use of elevated CO₂ and then highlighting the key research needs that could guide the development of more efficient and sustainable systems.

Variability in Physiological Responses Between Species and Environmental Conditions: Despite advances in the use of elevated CO₂ in protected systems, there is significant variability in the physiological responses of plants depending on species, variety, and environmental conditions. For example, crops like *Solanum curtilobum* and *Betula pubescens* have shown clear improvements in biomass and yield, while other species, such as *Castanea sativa*, exhibit transient or inconsistent responses. This behavior highlights the need to understand the factors that limit the stability of the long-term benefits of elevated CO₂, particularly under abiotic stress scenarios, such as high temperatures, UV-B radiation, or ozone pollution.

Effects on Nutritional Quality and Chemical Composition of Plants: Although CO₂ enrichment increases biomass in many crops, its impact on nutritional quality and chemical composition remains a concern. For example, in leafy green crops and medicinal plants, reductions in nitrogen concentration, chlorophyll levels, and certain bioactive compounds have been reported under supplemented CO₂ conditions. This suggests that the benefits in terms of yield may be accompanied by trade-offs in quality, which poses a challenge in meeting both nutritional and commercial demands in sustainable agricultural systems.

Limitations in Modeling and the Transfer of Results to Field Scenarios: Although CFD-based models and other computational tools have improved the prediction of CO₂ behavior in greenhouses, these models are often limited to simulations under ideal or semi-closed conditions. The lack of empirical data to validate these models in large-scale agricultural systems, particularly in perennial or long-cycle crops, makes it difficult to extrapolate findings to real-world scenarios. Additionally, interactions between CO₂

and other factors, such as nutrient dynamics and microbial networks in the soil, are underrepresented in current models.

Sustainability and Energy Efficiency in CO₂ Enrichment: CO₂ enrichment in greenhouses and plant factories involves high energy consumption, particularly in systems that require continuous light supplementation, controlled ventilation, and heating. While advances have been made in CO₂ use efficiency for crops like peppers and strawberries, the associated energy costs and carbon emissions limit the economic and environmental feasibility of these technologies. There is a critical gap in developing strategies that integrate renewable energy sources and CO₂ recycling methods to reduce environmental impact and enhance the sustainability of these systems.

5. Future Research Development Needs

Long-Term Responses in Perennial Crops: Current studies focus on short-cycle crops, leaving a significant gap in our understanding of the responses to elevated CO₂ in multi-annual and perennial crops, such as fruit trees or long-lived ornamental plants. Future research should explore how these species respond to elevated CO₂ over multiple growth cycles and how factors like biomass partitioning, longevity, and the quality of the final product are affected.

Integration of Multivariable Factors in Simulation Models: To address the complexity of protected agricultural systems, it is essential to develop robust models that integrate multiple variables, such as interactions between CO₂, nutrients, water, and soil microorganisms. These models should be capable of predicting not only plant growth and development but also the associated economic and environmental impacts, enabling the optimization of crop management based on specific conditions.

Innovations in Energy Management and Sustainability: It is essential to develop technologies that reduce dependence on fossil fuels in CO₂ enrichment systems. This includes the use of renewable energy sources, such as solar or wind power, as well as the development of integrated systems that allow for the efficient recycling of CO₂ within the same greenhouse. Research focused on circular economy principles in these systems could significantly enhance their economic and environmental viability.

Socioeconomic Impacts and Scalability of CO₂ Enrichment Technologies: While the benefits of elevated CO₂ in controlled systems are evident, large-scale implementation faces technical and socioeconomic challenges. It is crucial to assess the adaptation of these technologies to the needs of different agricultural regions considering infrastructure limitations, available resources, and initial costs. Additionally, exploring how to integrate these practices into sustainable agricultural policies that support global food security is essential. Focusing research on these areas will help overcome current challenges and maximize the potential of CO₂ enrichment as a key tool for sustainable and resilient agriculture.

6. Conclusions

CO₂ enrichment in protected agriculture is a promising strategy for improving crop productivity, resource use efficiency, and water conservation. However, its environmental impact depends on the CO₂ source and the energy matrix used. The adoption of carbon capture and reuse technologies, along with a transition to renewable energy sources, are crucial for maximizing benefits without increasing the carbon footprint.

This strategy has demonstrated significant improvements in biomass production and yield across various crops, although its effectiveness varies by species, phenological stage, and environmental conditions. Key challenges include ventilation, nutrient availability, and interactions with other environmental stressors, which must be addressed through advanced monitoring and optimized dosing strategies.

Technological advances in CO₂ injection and enrichment have enhanced photosynthetic efficiency and crop performance, but implementation must consider CO₂ sources, energy costs, and integration with environmental control systems. The use of liquefied CO₂, biogas, and carbon capture presents opportunities and challenges that require careful evaluation to ensure sustainability and profitability.

Modeling and simulation tools, such as CFD, neural networks, and physiological models, play a crucial role in optimizing CO₂ distribution, reducing resource consumption, and improving productivity. However, their effectiveness depends on experimental validation and integration with real-time control systems.

Plant Factories with Artificial Lighting (PFALs) and CO₂ enrichment offer high-yield solutions for controlled environment agriculture, yet their feasibility depends on energy efficiency and renewable integration. The development of precise regulation strategies and clean energy adoption could position them as a sustainable alternative for future agriculture.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/su17072809/s1>: PRISMA Protocol for Systematic Review. Reference [180] is cited in the supplementary materials.

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